# REGULARITY OF PLURISUBHARMONIC EXHAUSTIONS AND RELATED PROBLEMS

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Abstract. We construct Lelong-pseudoconvex but not weakly 1-complete threefolds, answering negatively Nakano's question. We show the affirmative side for divisor complements in compact Kähler surfaces. For total spaces of dual line bundles, a regularity criterion is shown for psh exhaustions, alongside the seminegative converse due to Ohsawa. We also generalize Vîjîitu's holomorphic convexity criterion and give a higher-dimensional analogue of Ohsawa's criterion.

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## 1. Introduction to the main results and applications

A *continuous plurisubharmonic (psh) exhaustion* of a reduced complex space $X$ is a continuous function $\rho : X \to \mathbb{R}$ whose pullback along every holomorphic disc is subharmonic and whose sublevel sets $\{\rho \leq c\}$ are compact for all $c \in \mathbb{R}$. We call $X$ *weakly* 1*-complete*, or *weakly pseudoconvex*, if it admits a smooth psh exhaustion.

Nakano asked in 1980 [Wak80, §I.1, Problem 1, p. 162] whether *Lelong-pseudoconvex* manifolds are weakly 1-complete, where Lelong-pseudoconvexity is defined by admitting a continuous psh exhaustion. Notably, in Demailly's thesis [Dem82b, Introduction, §1, p. 3], it allows any psh exhaustion in the definition of weak pseudoconvexity. On a Kähler manifold, a continuous psh exhaustion already gives a complete Kähler metric, as needed for his $L^2$ estimates for solving $\bar{\partial} u = f$ [Dem82a, Theorems 1.3 and 4.1].

### 1.1. Main Results.

In the following theorem we construct a quasi-projective counterexample to Nakano's question.

**Theorem 1.1.** *There exists a connected smooth quasi-projective Kähler threefold $M$ admitting a psh exhaustion of class $C^{0,1}_{\mathrm{loc}}$ but no psh exhaustion of class $C^{1,0}_{\mathrm{loc}}$.*

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Here by $C^{k,\alpha}_{\mathrm{loc}}$, for $k \geq 0$ an integer and $0 \leq \alpha \leq 1$, we mean $C^k$ functions whose derivatives of order $k$ are locally $\alpha$-Hölder in real coordinates. Thus $C^{0,1}_{\mathrm{loc}}$ means locally Lipschitz. At Hölder-exponent zero we also write $C^k = C^{k,0}_{\mathrm{loc}}$, because continuous derivatives are bounded on compact subsets.

The construction uses Grauert's ruled-surface example [Gra62, §4, 8(d)] and Koike's variant [Koi21, Example 4.1]. In each case the ruled surface $S$ has a negative section $Y$ and a distinguished line bundle $L$. This bundle is nef and big (Proposition 2.1) and admits a locally Lipschitz semipositive metric (Proposition 2.6). By Proposition 2.3, no neighborhood of $Y$ carries a $C^1$ semipositive metric on $L$. A Hermitian metric $h$ is semipositive if its weights $\varphi = -\log \|e\|^2_h$ in local holomorphic frames $e$ are psh. Unless otherwise specified, semipositive line bundles are understood to admit a smooth semipositive Hermitian metric. And we explicitly use the expression "continuous semipositive Hermitian metric", when only continuity is assumed. For a regularity class $\mathcal{R}$, a semipositive Hermitian metric of class $\mathcal{R}$ means a metric whose local weights $\varphi$ are real-valued plurisubharmonic functions of class $\mathcal{R}$. For nonsmooth metrics, curvature inequalities are understood in the sense of currents. We take $M = \mathrm{Tot}(L^{-1})$, the total space of the dual line bundle. By the regularity criterion (Theorem 3.4),

$$\begin{aligned} &M \text{ has a psh exhaustion of class } \mathcal{R} \\ \iff\quad &L \text{ has a semipositive metric of class } \mathcal{R}. \end{aligned}$$

This equivalence holds for every holomorphic line bundle over a compact complex manifold and every $C^r$, local Hölder, or real analytic ($C^\omega$) class $\mathcal{R}$ in (3.1). The metric obstruction on $L$ therefore rules out every $C^1$ psh exhaustion of $M$.

Taking the product of $M$ with a Hopf surface $H$ gives a five-dimensional non-Kähler counterexample. Since $H$ is compact, projection to $M$ pulls its locally Lipschitz psh exhaustion back to an exhaustion of $M \times H$. A $C^1$ psh exhaustion would restrict to one on the closed slice $M \times \{h\}$, and a Kähler form would restrict to one on $\{m\} \times H$. To obtain a non-Kähler counterexample in dimension three, we use a modification with a compact center.

**Theorem 1.2.** *Let $M$ be the threefold in Theorem 1.1. For every $x \in M$ there is a smooth non-Kähler threefold $N$ and a proper holomorphic modification $\mu : N \to M$ such that*

$$N \setminus \mu^{-1}(x) \simeq M \setminus \{x\}.$$

*The threefold $N$ admits a psh exhaustion of class $C^{0,1}_{\mathrm{loc}}$ but no psh exhaustion of class $C^{1,0}_{loc}$.*

We first blow up $x$ and choose a line and a conic in the exceptional plane. Following Hironaka's construction [Bar15, §6], we blow up these curves in opposite orders near their two intersection points. Two exceptional rational curves then have homology classes whose sum is zero, which rules out a Kähler form on $N$. The locally Lipschitz psh exhaustion of $M$ pulls back to $N$. Conversely, a $C^1$ psh exhaustion of $N$ could be made constant near $\mu^{-1}(x)$ by composing it with a smooth convex nondecreasing function tending to infinity. It would then descend to a $C^1$ psh exhaustion of $M$ (Proposition 4.1).

It remains open whether every Lelong-pseudoconvex complex manifold of dimension two is weakly 1-complete. The next theorem rules out counterexamples among complements of divisors in compact Kähler surfaces and, in particular, among smooth quasi-projective surfaces.

**Theorem 1.3.** *Let $S$ be a smooth compact connected Kähler surface and $D \neq 0$ an effective divisor. If $S \setminus \operatorname{supp} D$ admits a continuous psh exhaustion, it admits a smooth one. Consequently, the same implication holds for every smooth complex quasi-projective surface.*

Our proof strategy is to construct a smooth exhaustion of $X = S \setminus \operatorname{supp} D$ in two cases. If $X$ is a modification of a Stein space, we pull back a smooth strictly psh exhaustion from that

space. Otherwise we average positive $\partial\bar\partial$-closed currents supported on levels of an exhaustion. The resulting current has zero Lelong numbers and compact support in $X$. Using the current theorem of Dinh–Nguyên–Sibony [DNS22] and the Hodge index theorem, we show that there exists a divisor $A = \sum_j a_j D_j$ with positive integer coefficients such that

$$\operatorname{supp} A = \operatorname{supp} D, \qquad A^2 = 0, \qquad \mathcal{O}_S(A) \text{ is unitary flat near } D.$$

Let $s_A$ be the canonical section of $\mathcal{O}_S(A)$ and $h$ a flat Hermitian metric on this bundle near $\operatorname{supp} D$. The function $-\log \|s_A\|_h^2$ is pluriharmonic on a punctured neighborhood of $\operatorname{supp} D$ and tends uniformly to $+\infty$ on approach to $\operatorname{supp} D$. This is the *pluriharmonic logarithmic collar* from which we construct the smooth exhaustion. Section 5 also gives separate numerical and neighborhood criteria for existence of an exhaustion.

For a single smooth connected boundary curve $C \subset S$, the self-intersection $C^2 = \deg N_{C/S}$ and the holomorphic functions on $X = S \setminus C$ are related by

$$C^2 < 0 \quad \Longrightarrow \quad \mathcal{O}(X) = \mathbb{C} \quad \Longrightarrow \quad C^2 \le 0.$$

Indeed, when $C^2 < 0$, Grauert's contraction criterion [Gra62, §4, 8(e)] identifies $X$ with the complement of a point in a compact normal surface, and every holomorphic function extends across that point. When $C^2 > 0$, the curve $C$ is nef and $(K_S - mC)\cdot C < 0$ for $m$ large. Serre duality and Riemann–Roch then give $h^0(S, \mathcal{O}_S(mC)) > 1$. Dividing a section independent of $s_C^m$ by $s_C^m$ gives a nonconstant holomorphic function on $X$.

Ohsawa proved that a connected weakly 1-complete noncompact smooth surface is holomorphically convex if and only if it has a nonconstant holomorphic function [Ohs81, Proposition 1.4, p. 156]. Vîjîitu obtained a higher-dimensional result under a condition on the common level sets of global holomorphic functions [Vîj22b, Theorem 1.1]. Using [Vîj22b, Corollary 2.5], we obtain the following statement for maps to arbitrary Stein spaces, with no pure dimension requirements.

**Theorem 1.4.** *Let $X$ be a reduced complex space admitting a continuous psh exhaustion, and let $F : X \to S$ be a holomorphic map to a Stein space. Suppose that every irreducible component of every reduced fiber of $F$ is either compact or Stein. Then $X$ is holomorphically convex.*

The theorem allows both compact and noncompact fiber components of arbitrary dimension. It recovers [Vîj22b, Theorem 1.1] by means of [Vîj22b, Theorem 3.1]; see Corollary 6.3. Its hypotheses also cover spaces excluded by the singular-locus and rank assumptions of that result; see Remark 6.4. For a complement $S \setminus C$ admitting a continuous psh exhaustion, a nonconstant holomorphic function has one-dimensional fibers, so the theorem gives holomorphic convexity, in particular when $C^2 > 0$.

Throughout this paper, *regularization* means finding a smooth psh exhaustion on a space that admits a continuous one. Among smooth quasi-projective manifolds, Theorems 1.1 and 1.3 place the least possible counterexample dimension at three.

### 1.2. Applications to earlier problems.

**(i) Nakano's question.** As recalled above, Nakano asked whether every Lelong-pseudoconvex manifold is weakly 1-complete. Choose a locally Lipschitz psh exhaustion $\rho \ge 0$ on the threefold $M$ of Theorem 1.1. For each $n > 3$, the product $M \times \mathbb{C}^{n-3}$ is quasi-projective and admits the locally Lipschitz psh exhaustion

$$\rho_n(x, z) = \rho(x) + \|z\|^2.$$

A $C^1$ psh exhaustion on this product would restrict to one on the closed slice $M \times \{0\}$, contradicting Theorem 1.1. Thus the answer is negative in every dimension at least three.

Theorem 1.3 gives an affirmative answer for surfaces with a divisorial compactification in a compact Kähler surface, including smooth quasi-projective surfaces.

**(ii) Filip–Tosatti and the bounded-potential question.** Filip and Tosatti asked to prove that the distinguished nef and big line bundle in Grauert's ruled-surface example is not smoothly semipositive [FT18, Problem 2.2, p. 2989]. Zhang proved this for the original example using Koike's first obstruction class [Zha25, Theorem 1.2]. He also recorded a question suggested by Tosatti: for the original or modified Grauert example, does $c_1(L)$ contain a closed positive current with bounded local potentials and minimal singularities [Zha25, §4]? Proposition 2.6 and Remark 2.7 answer this affirmatively for both examples. The metric has locally Lipschitz weights and is smooth with strictly positive curvature on $S \setminus Y$. Its curvature has bounded local potentials and minimal singularities. In the notation of (2.2), the original and modified examples have

$$(Y^2, L^2) = (-1, 1), \qquad (Y^2, L^2) = (-2, 2),$$

respectively.

**(iii) Koike's first obstruction at $C^1$ regularity.** Koike extended Ueda's obstruction theory to line bundles. His Theorem 1.4 excludes smooth semipositivity when the first obstruction class is nonzero, and his Example 4.1 applies this to the modified Grauert example [Koi21]. For the Grauert triples $(Y, S, L)$ considered here, the nonzero extension class $\xi$ in (2.1) gives

$$u_1(Y, S, L) = -\xi \neq 0 \quad \text{in } H^1(Y, N_{Y/S}^{-1}) \simeq H^1(B, F).$$

Our Proposition 2.3 shows that no neighborhood of $Y$ carries a $C^1$ Hermitian metric on $L$ with psh weights; the same holds for every positive tensor power of $L$.

**(iv) Ohsawa–Vîjîitu convexity.** As stated by Ohsawa, the same method in [Ohs81] can also deduce that a connected noncompact smooth complex surface admitting a continuous psh exhaustion is holomorphically convex if and only if it has a nonconstant holomorphic function. Vîjîitu established a version for irreducible surfaces allowing noncontinuous psh exhaustions [Vîj22a, Theorem 1.1], and a higher-dimensional theorem assuming that the connected components of the singular locus are compact and that $\operatorname{rank}_x \mathcal{O}(X) = \dim X - 1$ at smooth points outside a compact set [Vîj22b, Theorem 1.1]. Our Theorem 1.4 extends the scope of this latter statement to maps to Stein spaces whose fiber components are compact or Stein. Its proof uses [Vîj22b, Corollary 2.5], and Corollary 6.3 recovers [Vîj22b, Theorem 1.1]. The products $C \times \mathbb{C}^2$, for a projective nodal cubic $C$, and $\mathbf{P}^2 \times \mathbb{C}$ satisfy Theorem 1.4 while failing, respectively, the singular-locus and rank hypotheses of [Vîj22b, Theorem 1.1]; see Remark 6.4.

**(v) Ohsawa's seminegative converse.** Ohsawa observed that the total space of a seminegative holomorphic line bundle over a compact complex manifold is weakly 1-complete and discussed the converse [Ohs21, §6, Proposition 6.1 and the following paragraph]. Ohsawa's Theorem 6.1 treats affine line bundles over compact Riemann surfaces. For linear line bundles, taking $\mathcal{R} = C^\infty$ in Theorem 3.4 proves the converse over any compact complex manifold. The criterion also preserves the $C^r$, local Hölder, and real analytic classes in (3.1).

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## 2. Distinguished line bundles on Grauert's ruled surface

The two regularity properties of the distinguished bundle $L$ come from its behavior along the negative section $Y$. A $C^1$ semipositive metric would split the extension defining the ruled surface. On the other hand, the negativity of $Y$ allows us to construct a semipositive metric whose weights are locally Lipschitz across $Y$.

2.1. **Grauert's construction and its line bundle.** Let $B$ be a smooth connected projective curve of genus $g \geq 2$, let $F$ be a line bundle of degree $d > 0$, and choose $0 \neq \xi \in H^1(B, F)$. Let

$$0 \longrightarrow F \longrightarrow E \longrightarrow \mathcal{O}_B \longrightarrow 0 \tag{2.1}$$

be a non-split extension with class $\xi$. We write $\mathbf{P}(E) = \mathbf{P}_{\text{lines}}(E)$ for the bundle of lines in $E$, with tautological line bundle $\mathcal{O}_{\mathbf{P}(E)}(-1)$. Set

$$S = \mathbf{P}(E), \qquad \pi : S \to B, \qquad Y = \mathbf{P}(F), \qquad \boxed{L = \mathcal{O}_S(Y) \otimes \pi^* F}. \tag{2.2}$$

We identify the section $Y$ with $B$ via $\pi$. By GAGA, $E$ is algebraic and $S$ is a smooth projective surface. The choice $F = K_B$ works for every $g \geq 2$, since Serre duality gives

$$\deg K_B = 2g - 2 > 0, \qquad H^1(B, K_B) \simeq H^0(B, \mathcal{O}_B)^* \simeq \mathbb{C}.$$

In genus two we may also take $F = \mathcal{O}_B(p)$ for $p \in B$. A nonconstant meromorphic function with at most a simple pole at $p$ would give a degree-one map $B \to \mathbf{P}^1$. Thus $h^0(B, \mathcal{O}_B(p)) = 1$, and Riemann–Roch gives $h^1(B, \mathcal{O}_B(p)) = 1$ as well. These two choices of $F$ recover Grauert's example for $F = \mathcal{O}_B(p)$ [Gra62, §4, 8(d)] and Koike's variant for $F = K_B$ [Koi21, Example 4.1].

Choose local frames $f_j$ of $F$ on $U_j \subset B$ and lifts $v_j$ of $1 \in \mathcal{O}_B$, with

$$f_k = g_{jk} f_j, \qquad v_k = v_j + c_{jk} f_j.$$

The cocycle $\{v_k - v_j = c_{jk} f_j\}$ represents $\xi$. On $S \setminus Y$, write a line as $[t_j f_j + v_j]$. Near $Y$, write it as $[f_j + w_j v_j]$, with $w_j = 1/t_j$ where both coordinates are defined. Comparing coefficients on overlaps gives

$$t_j = g_{jk} t_k + c_{jk}, \qquad w_j = \frac{w_k}{g_{jk} + c_{jk} w_k}, \qquad w_k = \frac{g_{jk} w_j}{1 - c_{jk} w_j}. \tag{2.3}$$

The quotient map $E \to \mathcal{O}_B$ induces a section of $\mathcal{O}_S(1)$ whose zero divisor is $Y$. Consequently

$$\begin{aligned} \mathcal{O}_S(Y) &\simeq \mathcal{O}_S(1), \qquad N_{Y/S} \simeq \operatorname{Hom}(F, E/F) \simeq F^{-1}, \\ Y^2 &= -d, \qquad L|_Y \simeq F^{-1} \otimes F \simeq \mathcal{O}_Y. \end{aligned} \tag{2.4}$$

Let $s_Y$ be the canonical section of $\mathcal{O}_S(Y)$. Near $Y$, let $d_j$ be the frame defined by $s_Y = w_j d_j$, and put $\ell_j = d_j \otimes \pi^* f_j$. Then

$$d_k = \frac{1 - c_{jk} w_j}{g_{jk}} d_j, \qquad \ell_k = (1 - c_{jk} w_j) \ell_j. \tag{2.5}$$

In particular, the frames $\ell_j|_Y$ agree and give the trivialization in (2.4).

**Proposition 2.1.** *The line bundle $L$ as per (2.2) is nef and big, with*

$$L^2 = d, \qquad L \cdot Y = 0, \qquad L \cdot C > 0,$$

*for every irreducible curve $C \neq Y \subset S$. The bundle $A = L^{\otimes 2} \otimes \mathcal{O}_S(-Y)$ is ample. Every section of a positive tensor power of $L$ vanishes along $Y$. In particular, $L$ is not semiample.*

*Proof.* Let $P$ be a fiber of $\pi$. The intersection identities are

$$Y^2 = -d, \quad Y \cdot \pi^*F = d, \quad (\pi^*F)^2 = 0, \quad Y \cdot P = 1, \quad \pi^*F \cdot P = 0.$$

Thus $L^2 = d$, $L \cdot Y = 0$, and $L \cdot P = 1$. For an irreducible horizontal curve $D \neq Y$, let $n = \deg(\pi|_{\widetilde{D}}) > 0$, where $\widetilde{D}$ is its normalization. Then

$$L \cdot D = Y \cdot D + nd > 0.$$

Thus $L$ is nef, and $L^2 = d > 0$ makes it big by the numerical criterion for nef bundles on a projective surface. The only curve on which $L$ has degree zero is $Y$. For $A$ we have

$$A = \mathcal{O}_S(Y) \otimes \pi^*(F^{\otimes 2}), \qquad A^2 = 3d,$$
$$A \cdot Y = d, \qquad A \cdot P = 1, \qquad A \cdot D = Y \cdot D + 2nd > 0.$$

The Nakai–Moishezon criterion now gives ampleness.

To prove the assertion about sections, we show that $S \setminus Y$ contains no compact irreducible curve. Such a curve $D$ cannot be contained in a fiber of $\pi$, since each fiber of $\pi|_{S\setminus Y}$ is an affine line and hence contains no compact curve. Thus the restriction $\pi|_D : D \to B$ is nonconstant and, since $D$ is compact, surjective. Its normalization would give a finite map $p : \widetilde{D} \to B$ of degree $n > 0$. Away from the branch values, put

$$a_j(b) = \frac{1}{n} \sum_{x \in p^{-1}(b)} t_j(x).$$

The sum is holomorphic and single-valued, and is locally bounded at branch values by compactness of the inverse image of a small closed disc. It extends holomorphically across them. Equation (2.3) implies

$$a_j = g_{jk} a_k + c_{jk}, \qquad c_{jk} f_j = a_j f_j - a_k f_k.$$

This makes the cocycle representing $\xi$ a coboundary, a contradiction.

Now let $\sigma \in H^0(S, L^{\otimes m})$, with $m > 0$. Since $L^{\otimes m}|_Y \simeq \mathcal{O}_Y$, the restriction $\sigma|_Y$ can be identified with a holomorphic function on the compact connected curve $Y$, and is therefore constant. Suppose this constant is nonzero. Then $\sigma$ is nowhere zero on $Y$, so its zero divisor is disjoint from $Y$. For any fiber $P$ of $\pi$, the section $\sigma|_P$ is nonzero at the point $P \cap Y$ and hence is not identically zero. Since

$$L^{\otimes m}|_P \simeq \mathcal{O}_{\mathbf{P}^1}(m),$$

this section has exactly $m > 0$ zeros, counted with multiplicity. Thus the zero divisor of $\sigma$ is nonempty, and any irreducible component of its support is a compact irreducible curve in $S \setminus Y$, contradicting what we have just proved. Hence $\sigma|_Y = 0$, as claimed. □

2.2. **The first normal derivative and the $C^1$ obstruction.** On $Y$, the weights of a semipositive metric on $L$ glue to a subharmonic function and hence are constant in the frames $\ell_j$. The next lemma determines what plurisubharmonicity then imposes on their first derivatives normal to $Y$.

**Lemma 2.2** (First normal derivative)**.** *Let $D \subset \mathbb{C}$ be a disc and let $\varphi$ be a real-valued $C^1$ psh function on a neighborhood of $D \times \{0\}$ in $\mathbb{C}^2$. If $\varphi(z, 0)$ is constant, then $b(z) = \partial_w \varphi(z, 0)$ is holomorphic.*

*Proof.* After subtracting a constant, we may assume that $\varphi(z, 0) = 0$. Since $\varphi$ is $C^1$,

$$\varphi(z, w) = 2\,\mathrm{Re}(b(z)w) + o(|w|), \tag{2.6}$$

uniformly for $z$ in compact subsets of $D$. Let $f$ be holomorphic on $D$. On each relatively compact subdisc, the functions

$$u_t^{\pm}(z) = \frac{\varphi(z, \pm t f(z))}{t}$$

are subharmonic for all sufficiently small $t > 0$, since $\varphi$ is psh. By (2.6), they converge locally uniformly as $t \to 0^+$ to $\pm 2\,\mathrm{Re}(bf)$. Both limits are subharmonic, so $\mathrm{Re}(bf)$ is harmonic. Taking $f = 1, \sqrt{-1}$ shows that the real and imaginary parts of $b$ are harmonic; in particular, $b$ is smooth. Taking $f = z, \sqrt{-1}z$ likewise shows that $zb$ is harmonic. Hence

$$0 = \Delta(zb) = z\Delta b + 4\partial_{\bar{z}} b = 4\partial_{\bar{z}} b.$$

Therefore $b$ is holomorphic. □

**Proposition 2.3.** *No neighborhood of $Y$ carries a $C^1$ Hermitian metric on $L$ with psh weights. The same holds for $L^{\otimes m}$, $m \geq 1$.*

*Proof.* Recall that the weights $\varphi_j = -\log \|\ell_j\|_h^2$ of a Hermitian metric $h$ on a line bundle satisfy

$$\varphi_k = \varphi_j - \log|\gamma_{jk}|^2 \qquad \text{when } \ell_k = \gamma_{jk}\ell_j. \tag{2.7}$$

With $d^c = \frac{\sqrt{-1}}{4\pi}(\bar{\partial} - \partial)$ and $dd^c = \frac{\sqrt{-1}}{2\pi}\partial\bar{\partial}$, the curvature is $\Theta_h = dd^c\varphi_j$, understood in the sense of currents when the weights are not $C^2$.

Suppose that $L$ admits a $C^1$ Hermitian metric with psh weights $\varphi_j$ in the frames $\ell_j$ with $\gamma_{jk} = 1 - c_{jk}w_j$ near $Y$. By (2.7) and (2.5),

$$\varphi_k(z, w_k) = \varphi_j(z, w_j) - \log|1 - c_{jk}w_j|^2. \tag{2.8}$$

Since the frames $\ell_j|_Y$ agree, the restrictions $\varphi_j|_Y$ define a global subharmonic function on the compact connected curve $Y$, and this function is constant. Lemma 2.2 therefore shows that

$$b_j(z) = \partial_{w_j}\varphi_j(z, 0)$$

is holomorphic on $U_j$. By (2.3), $\partial w_k/\partial w_j = g_{jk}$ along $Y$. Differentiating (2.8) with respect to $w_j$ at $w_j = 0$ thus gives

$$g_{jk}b_k = b_j + c_{jk}.$$

Consequently,

$$v_k - b_k f_k = v_j + (c_{jk} - g_{jk}b_k)f_j = v_j - b_j f_j.$$

Thus the local holomorphic sections $v_j - b_j f_j$ glue to a global section $s$ of $E$ whose image under $E \to \mathcal{O}_B$ is 1. The map $a \mapsto as$ is then a holomorphic right inverse of this quotient map, splitting the extension (2.1) and contradicting $\xi \neq 0$.

Finally, a $C^1$ semipositive metric on $L^{\otimes m}$ would induce one with the same regularity on $L$: its weights in the frames $\ell_j^{\otimes m}$, divided by $m$, satisfy the transition rule for weights on $L$ and remain $C^1$ and psh. This proves the assertion for every $m \geq 1$. □

Under the identification $N_{Y/S}^{-1} \simeq F$, the local generator $dw_j|_Y$ corresponds to $f_j$. By the definition of Koike's first obstruction class [Koi21, §3.2, p. 2247] and (2.5),

$$u_1(Y, S, L) = [\{-c_{jk}\,dw_j\}] = -\xi \quad \text{in } H^1(Y, N_{Y/S}^{-1}) \simeq H^1(B, F).$$

Koike's obstruction theorem applies to smooth weights [Koi21, Theorem 1.4]. Wu also computes the smooth obstruction in Grauert's example [Wu26, Example 3.7]. Our Lemma 2.2 therewithal gives a more precise obstruction at $C^1$ regularity.

2.3. **From local to global Lipschitz metrics.** To construct the $C^{0,1}_{loc}$-semipositive metric, we add a multiple of the norm of the defining section to a smooth local weight. Away from the curve this term supplies positive curvature in both the tangential and normal directions.

**Lemma 2.4** (Local Lipschitz metric)**.** *Let $D$ be a compact smooth connected curve in a complex surface $X$, with $D^2 < 0$, and let $H$ be a holomorphic line bundle near $D$ such that $H|_D$ admits a smooth flat Hermitian metric. Then there exist a neighborhood $U$ of $D$, smooth Hermitian metrics $h_0$ on $H|_U$ and $h_D$ on $\mathcal{O}_X(D)|_U$, and a constant $\Lambda > 0$ such that*

$$h_{\mathrm{loc}} = h_0 e^{-\Lambda s}, \qquad s = \|s_D\|_{h_D},$$

*has locally Lipschitz psh weights on $U$ and smooth strictly psh weights on $U \setminus D$. Here $s_D$ is the canonical section of $\mathcal{O}_X(D)$ vanishing along $D$.*

*Proof.* Extend the flat metric on $H|_D$ to a smooth metric $h_0$ near $D$, and put $\theta = \Theta_{h_0}(H)$. Then $\theta|_{TD} = 0$. Since $\deg N_{D/X} = D^2 < 0$, the normal bundle admits a smooth metric with strictly negative curvature. Indeed, the curvature of any initial metric can be adjusted to a negative area form of the same integral by solving a Poisson equation on $D$. Using $N_{D/X} \simeq \mathcal{O}_X(D)|_D$, extend this metric to a smooth metric $h_D$ on $\mathcal{O}_X(D)$ near $D$. Set

$$\eta = -\Theta_{h_D}(\mathcal{O}_X(D)), \qquad q = \|s_D\|^2_{h_D}, \qquad s = q^{1/2}.$$

Thus $\eta|_{TD} > 0$.

Choose local coordinates $(z, w)$ with $D = \{w = 0\}$ and a frame $d$ of $\mathcal{O}_X(D)$ such that $s_D = wd$. Writing $\|d\|^2_{h_D} = e^{\beta}$, we have

$$q = |w|^2 e^{\beta}, \qquad s = |w| e^{\beta/2}, \qquad \eta = dd^c\beta. \tag{2.9}$$

After shrinking a finite collection of relatively compact charts covering $D$, we may assume that $|w\beta_w| \le 1/2$. Consider the smooth $(1,0)$-frame

$$V_1 = \partial_z - \frac{w\beta_z}{1 + w\beta_w}\partial_w, \qquad V_2 = \partial_w.$$

It satisfies $V_1|_D = \partial_z$ and, away from $D$,

$$\partial \log q(V_1) = 0, \qquad \partial \log q(V_2) = \frac{1}{w} + \beta_w.$$

Since $s = \exp(\frac{1}{2}\log q)$, direct differentiation gives

$$dd^c s = \frac{s}{2}\eta + \frac{s}{4}\frac{\sqrt{-1}}{2\pi}\partial \log q \wedge \bar\partial \log q \qquad \text{on } U \setminus D. \tag{2.10}$$

We write the coefficients of these forms in the frame $(V_1, V_2)$, omitting the common factor $\sqrt{-1}/(2\pi)$. The preceding identities yield

$$(dd^c s)_{1\bar 1} = \frac{s}{2}\eta_{1\bar 1}, \qquad (dd^c s)_{1\bar 2} = \frac{s}{2}\eta_{1\bar 2},$$
$$(dd^c s)_{2\bar 2} = \frac{s}{2}\eta_{2\bar 2} + \frac{s}{4}|w^{-1} + \beta_w|^2.$$

These are tensor coefficients in a smooth frame; the frame need not be holomorphic.

Since $s$ is comparable to $|w|$ and $|1 + w\beta_w| \ge 1/2$, the last term is bounded below by a positive constant times $s^{-1}$. Moreover, $\eta_{1\bar 1} > 0$ near $D$, while smoothness and $\theta_{1\bar 1}|_D = 0$ give $\theta_{1\bar 1} = O(s)$. Thus, after shrinking the neighborhood, there are constants $a, b, C > 0$, independent of the chart, such that

$$\begin{aligned} &(dd^c s)_{1\bar 1} \ge as, \qquad (dd^c s)_{2\bar 2} \ge b/s, \quad |(dd^c s)_{1\bar 2}| \le Cs, \\ &\theta_{1\bar 1} \ge -Cs, \qquad \theta_{2\bar 2} \ge -C, \qquad |\theta_{1\bar 2}| \le C. \end{aligned} \tag{2.11}$$

Choose $\Lambda > 0$ sufficiently large that

$$\Lambda a - C > 0, \qquad (\Lambda a - C)\frac{\Lambda b}{2} > (C+1)^2. \tag{2.12}$$

Then shrink $U$ so that $Cs \le \Lambda b/2$ and $\Lambda C s \le 1$. For $T = \theta + \Lambda dd^c s$, we obtain

$$T_{1\bar{1}} \ge (\Lambda a - C)s > 0, \qquad T_{2\bar{2}} \ge \frac{\Lambda b}{2s} > 0, \qquad |T_{1\bar{2}}| \le C+1.$$

Consequently,

$$\det T \ge (\Lambda a - C)\frac{\Lambda b}{2} - (C+1)^2 > 0,$$

so $T$ is strictly positive on $U \setminus D$.

If $\varphi_0$ is a local weight of $h_0$, then $\varphi_0 + \Lambda s$ is a weight of $h_{\mathrm{loc}}$. It is smooth and strictly psh off $D$, and continuous across $D$. The removable-singularity theorem for psh functions therefore makes it psh on all of $U$. Finally, $s = |w|e^{\beta/2}$ is locally Lipschitz, so these weights are locally Lipschitz as well. □

**Proposition 2.5** (Global Lipschitz metric)**.** *Let $X$ be a smooth projective surface, $D \subset X$ a smooth connected curve with $D^2 < 0$, and $H$ a line bundle such that $H|_D$ admits a smooth flat metric. Suppose*

$$A = H^{\otimes m} \otimes \mathcal{O}_X(-a_0 D)$$

*is ample for some integers $m, a_0 > 0$. Then $H$ admits a locally Lipschitz semipositive Hermitian metric, smooth with strictly positive curvature on $X \setminus D$.*

*Proof.* Let $U$, $h_0$, $h_D$, and $\Lambda$ be as in Lemma 2.4. After shrinking $U$, extend $h_0$ and $h_D$ to smooth metrics on $X$ without changing them on $U$. Put

$$q = \|s_D\|^2_{h_D}, \qquad s = q^{1/2}, \qquad \theta = \Theta_{h_0}(H), \qquad u_{\mathrm{loc}} = \Lambda s.$$

Thus $h_0 e^{-u_{\mathrm{loc}}}$ has locally Lipschitz psh weights on $U$, smooth and strictly psh on $U \setminus D$.

Since $A$ is ample, choose a smooth Hermitian metric $h_A$ with $\omega_A = \Theta_{h_A}(A) > 0$. The isomorphism $H^{\otimes m} \simeq A \otimes \mathcal{O}_X(a_0 D)$ defines a smooth real-valued function $G$ on $X$ by

$$h_A \otimes h_D^{a_0} = h_0^m e^{-G}.$$

Taking curvatures gives

$$m\theta + dd^c G = \omega_A + a_0 \Theta_{h_D}(\mathcal{O}_X(D)).$$

The Poincaré–Lelong formula gives

$$dd^c \log q = [D] - \Theta_{h_D}(\mathcal{O}_X(D)), \tag{2.13}$$

where $[D]$ denotes the current of integration over $D$. Consequently, the function $\psi = (G + a_0 \log q)/m$ satisfies

$$\theta + dd^c \psi = \frac{1}{m}\omega_A + \frac{a_0}{m}[D] \tag{2.14}$$

in the sense of currents. In particular, $h_0 e^{-\psi}$ has smooth strictly psh weights on $X \setminus D$, and $\psi$ tends uniformly to $-\infty$ as one approaches $D$.

We join these two metrics using a regularized maximum. Fix $\varepsilon > 0$ and choose an even nonnegative smooth function $\chi_\varepsilon$ supported in $(-\varepsilon, \varepsilon)$ with integral one. Set

$$\rho_\varepsilon(t) = \int_{\mathbb{R}} |t-a|\chi_\varepsilon(a)\, da, \qquad \mathcal{M}_\varepsilon(x,y) = \frac{x + y + \rho_\varepsilon(x-y)}{2}.$$

Since $\rho_\varepsilon'' \geq 0$ and $|\rho_\varepsilon'| \leq 1$, the function $\mathcal{M}_\varepsilon$ is smooth, convex, and nondecreasing in each variable. Moreover,

$$\mathcal{M}_\varepsilon(x+c, y+c) = c + \mathcal{M}_\varepsilon(x,y),$$
$$\mathcal{M}_\varepsilon(x,y) = \max\{x,y\} \qquad \text{if } |x-y| \geq \varepsilon.$$

To see that it preserves strict plurisubharmonicity, write

$$\mathcal{L}_f(v) = \sum_{j,k} f_{z_j \bar{z}_k} v_j \overline{v_k}$$

for the Levi form of a smooth function $f$. Convexity and the chain rule give

$$\mathcal{L}_{\mathcal{M}_\varepsilon(p_1,p_2)}(v) \geq \sum_{j=1}^{2} (\mathcal{M}_\varepsilon)_j(p_1,p_2)\, \mathcal{L}_{p_j}(v).$$

The coefficients on the right are nonnegative and sum to one. Hence the regularized maximum is strictly psh whenever both $p_1$ and $p_2$ are smooth and strictly psh.

Since $X$ is compact and $q$ vanishes exactly on $D$, choose $b > 0$ such that $\{q \leq 2b\} \subset U$. We can then choose $N \in \mathbb{R}$ and $0 < a < b/4$ such that

$$\begin{aligned} \psi + N &> u_{\mathrm{loc}} + 2\varepsilon \quad \text{on } \{b/2 \leq q \leq b\}, \\ \psi + N &< u_{\mathrm{loc}} - 2\varepsilon \quad \text{on } \{0 < q \leq 2a\}. \end{aligned} \tag{2.15}$$

Indeed, first choose $N$ sufficiently large on the compact outer shell. With $N$ fixed, the second inequality follows for sufficiently small $a$, because $\psi \to -\infty$ uniformly near $D$ and $u_{\mathrm{loc}} \geq 0$.

Define

$$h = h_0 e^{-u}, \qquad u = \begin{cases} u_{\mathrm{loc}}, & q \leq a, \\ \mathcal{M}_\varepsilon(u_{\mathrm{loc}}, \psi + N), & a < q < b, \\ \psi + N, & q \geq b. \end{cases} \tag{2.16}$$

By (2.15), these expressions agree on neighborhoods of the joining levels $q = a$ and $q = b$. If $\varphi_0$ is a local weight of $h_0$, then on $\{a < q < b\}$ the corresponding weight of $h$ is

$$\varphi_0 + u = \mathcal{M}_\varepsilon(\varphi_0 + u_{\mathrm{loc}}, \varphi_0 + \psi + N),$$

by the translation identity for $\mathcal{M}_\varepsilon$. Both arguments are smooth and strictly psh there, so the resulting weight has the same properties. Near $D$, the metric agrees with the one from Lemma 2.4, while on $\{q \geq b\}$ its curvature is $\omega_A/m > 0$. Thus $h$ has locally Lipschitz psh weights on $X$ and is smooth with strictly positive curvature on $X \setminus D$. □

**Proposition 2.6.** *For the ruled surface (2.2), $L$ admits a locally Lipschitz semipositive Hermitian metric, smooth with strictly positive curvature on $S \setminus Y$. Its curvature has bounded local potentials and minimal singularities.*

*Proof.* Equation (2.4) and Proposition 2.1 give

$$Y^2 = -d < 0, \qquad L|_Y \simeq \mathcal{O}_Y, \qquad L^{\otimes 2} \otimes \mathcal{O}_S(-Y) \text{ ample.}$$

Apply Proposition 2.5 with $(X, D, H, m, a_0) = (S, Y, L, 2, 1)$. The resulting curvature current has locally bounded potentials and hence minimal singularities. □

*Remark* 2.7. For $g = 2$, the choices $F = \mathcal{O}_B(p)$ and $F = K_B$ yield, respectively,

$$(Y^2, L^2) = (-1, 1), \qquad (Y^2, L^2) = (-2, 2).$$

Proposition 2.6 gives bounded potentials with minimal singularities for both classes considered in [Zha25, Section 4]. Proposition 2.3 excludes $C^1$ semipositive weights in either case.

## 3. The regularity criterion

For a continuous semipositive Hermitian metric on $L$, the squared norm on the dual bundle gives a psh exhaustion. To recover a metric from an exhaustion, we first average over rotations of the fibers. A suitable sublevel set then meets each fiber in a disc whose radius determines the metric. We carry out both constructions for the local regularity classes

$$\begin{aligned}\mathcal{R} \in \{C^r : r \in \mathbb{Z}_{\geq 0} \cup \{\infty\}\} \cup \{C^\omega\} \\ \cup \{C^{k,\alpha}_{\mathrm{loc}} : k \in \mathbb{Z}_{\geq 0},\ 0 < \alpha \leq 1\}.\end{aligned} \tag{3.1}$$

For every class in (3.1), the positive metric coefficients $h_j$, their reciprocals, and the weights $-\log h_j$ have the same regularity.

**Lemma 3.1.** *Let $L \to X$ be a holomorphic line bundle over a compact complex manifold. If $h$ is a semipositive Hermitian metric of class $\mathcal{R}$, then $\rho(v) = \|v\|^2_{h^{-1}}$ is a psh exhaustion of $\mathrm{Tot}(L^{-1})$ of the same class.*

*Proof.* Let $e_j$ be a local frame of $L$ over $B_j \subset X$, with $\|e_j\|^2_h = e^{-\varphi_j}$. For $v = \lambda_j e_j^*$,

$$\rho(z, \lambda_j) = |\lambda_j|^2 e^{\varphi_j(z)}. \tag{3.2}$$

This formula gives the claimed regularity on the total space. Off the zero section, $\log \rho = \log |\lambda_j|^2 + \varphi_j$ is psh, so $\rho$ is psh there by convex increasing composition. Continuity and removability across the zero section give plurisubharmonicity everywhere.

Fix a smooth Hermitian metric $k$ on $L^{-1}$. The ratio of $h^{-1}$ to $k$ is a positive continuous function on the compact base. Its minimum $m > 0$ gives $\rho(v) \geq m\|v\|^2_k$, so each sublevel of $\rho$ is a closed subset of a compact disc bundle. □

**Lemma 3.2.** *If a compact group $G$ acts continuously on a locally compact Hausdorff space and $F_0$ is a continuous exhaustion, then $F(x) = \int_G F_0(gx)\, dg$, with normalized Haar measure $dg$, is an exhaustion. For the fiberwise circle action on a holomorphic line bundle, averaging preserves plurisubharmonicity and every class in (3.1).*

*Proof.* Continuity follows from uniform continuity on compact orbit sets. For $K_c = \{F_0 \leq c\}$,

$$\{F \leq c\} \subset G \cdot K_c.$$

Indeed, $x \notin G \cdot K_c$ implies $\min_{g \in G} F_0(gx) > c$, and therefore $F(x) > c$. The set $G \cdot K_c$ is the continuous image of the compact set $G \times K_c$, so its closed subset $\{F \leq c\}$ is compact.

For the circle action $R_\theta(v) = e^{\sqrt{-1}\theta} v$, put

$$F(v) = \frac{1}{2\pi} \int_0^{2\pi} F_0(R_\theta v)\, d\theta. \tag{3.3}$$

If $F_0$ is psh, integration of the submean inequality on each holomorphic disc gives the same inequality for $F$. Invariance under rotation follows from translation invariance of $d\theta$.

In real bundle coordinates $x = (z, \lambda)$, write $T_\theta x = (z, e^{\sqrt{-1}\theta}\lambda)$. The maps $T_\theta$ are linear with uniformly bounded norms. For $F_0 \in C^r$, $r \geq 1$, differentiation on compact coordinate sets yields

$$D^\beta F(x) = \frac{1}{2\pi} \int_0^{2\pi} D^\beta_x (F_0 \circ T_\theta)(x)\, d\theta, \qquad |\beta| \leq r.$$

For $F_0 \in C^{k,\alpha}_{\mathrm{loc}}$, a finite cover of the compact orbit set gives, uniformly in $\theta$,

$$|D^\beta (F_0 \circ T_\theta)(x) - D^\beta (F_0 \circ T_\theta)(y)| \leq C|x - y|^\alpha, \qquad |\beta| = k.$$

Integration gives the same estimate for $D^\beta F$.

For the analytic case, $\mathcal{G}(x,\theta) = F_0(T_\theta x)$ is jointly real analytic near the compact set $\{x_0\}\times S^1$. A finite cover by analytic coordinate neighborhoods and Cauchy's estimates give constants $M, R > 0$ such that

$$\left|\frac{D_x^\beta \mathcal{G}(x_0,\theta)}{\beta!}\right| \le MR^{-|\beta|} \qquad (\beta \in \mathbb{Z}^m_{\ge 0},\ \theta \in S^1), \tag{3.4}$$

where $m$ is the real coordinate dimension. The Taylor series in $x$ converges uniformly in $\theta$ on every smaller polydisc. Integrating term by term gives

$$F(x) = \sum_\beta \left(\frac{1}{2\pi}\int_0^{2\pi} \frac{D_x^\beta \mathcal{G}(x_0,\theta)}{\beta!}\, d\theta\right)(x-x_0)^\beta.$$

This convergent power series proves $F \in C^\omega$. □

**Proposition 3.3.** *Let $L \to X$ be a holomorphic line bundle over a compact complex manifold. If $\mathrm{Tot}(L^{-1})$ admits a psh exhaustion of class $\mathcal{R}$ in (3.1), then $L$ admits a semipositive Hermitian metric of the same class.*

*Proof.* Let $F$ be the circle average (3.3) of the given exhaustion. For a local frame $e_j$ of $L$ over $B_j$, define

$$U_j(z,t) = F(e^{t/2}e_j^*(z)), \qquad V_j(z,\zeta) = F(e^{\zeta/2}e_j^*(z)).$$

Both parametrizations are real analytic, so $U_j$ and $V_j$ inherit the class $\mathcal{R}$. The second parametrization is holomorphic, so $V_j$ is psh. Circle invariance gives

$$V_j(z,\zeta) = U_j(z,\operatorname{Re}\zeta), \qquad F(\lambda_j e_j^*(z)) = U_j(z,\log|\lambda_j|^2) \quad (\lambda_j \ne 0).$$

Fix $z$ and write $u(t) = U_j(z,t)$. Since $V_j$ is independent of $\operatorname{Im}\zeta$, its subharmonicity in $\zeta$ gives $u'' \ge 0$ in the sense of distributions. Thus $u$ is convex. Continuity at the zero section and properness of $F$ give

$$\lim_{t\to-\infty} u(t) = F(0_z), \qquad \lim_{t\to+\infty} u(t) = +\infty. \tag{3.5}$$

By convexity and its finite limit at $-\infty$, $u$ is nondecreasing and strictly increasing wherever $u > F(0_z)$. Hence, for $c > \max_{z\in X} F(0_z)$, there is a unique $\sigma_j(z)$ such that

$$U_j(z,\sigma_j(z)) = c, \qquad U_j(z,t) \le c \iff t \le \sigma_j(z). \tag{3.6}$$

In particular,

$$\{F < c\} \cap L^{-1}|_{B_j} = \{(z,\lambda_j) : |\lambda_j|^2 < e^{\sigma_j(z)}\}. \tag{3.7}$$

For $s_0 = \sigma_j(z_0)$ and $\varepsilon > 0$,

$$U_j(z_0, s_0-\varepsilon) < c < U_j(z_0, s_0+\varepsilon).$$

These strict inequalities persist near $z_0$ and force $s_0-\varepsilon < \sigma_j(z) < s_0+\varepsilon$. Thus $\sigma_j$ is continuous.

To prove that $-\sigma_j$ is psh, let $f$ be a holomorphic map from a neighborhood of $\overline{\Delta}$ to $B_j$, where $\Delta = \{\zeta\in\mathbb{C} : |\zeta| < 1\}$. Let $H$ be the harmonic extension of $\sigma_j \circ f|_{\partial\Delta}$, continuous on $\overline{\Delta}$. Since $\Delta$ is simply connected, there is a holomorphic $g$ with $\operatorname{Re} g = H$. The function

$$q(\zeta) = F\big(e^{g(\zeta)/2}e_j^*(f(\zeta))\big) = U_j(f(\zeta), H(\zeta))$$

is subharmonic on $\Delta$, and the second expression extends continuously to the boundary with value $c$. The maximum principle and (3.6) give

$$q \le c \implies H \le \sigma_j\circ f \implies -\sigma_j(f(0)) \le \frac{1}{2\pi}\int_0^{2\pi} -\sigma_j(f(e^{\sqrt{-1}\theta}))\, d\theta.$$

The submean inequality for every such disc proves that $-\sigma_j$ is psh.

If $e_k = g_{jk}e_j$, then $\lambda_k = g_{jk}\lambda_j$, and the radii of the same sublevel in (3.7) satisfy

$$e^{\sigma_k} = |g_{jk}|^2 e^{\sigma_j}, \qquad \sigma_k = \sigma_j + \log|g_{jk}|^2.$$

Thus

$$\psi_j = -\sigma_j, \qquad \psi_k = \psi_j - \log|g_{jk}|^2, \qquad \|e_j\|_h^2 = e^{-\psi_j} = e^{\sigma_j} \tag{3.8}$$

defines a continuous Hermitian metric on $L$ with $\Theta_h(L) = dd^c\psi_j \geq 0$.

We now recover the regularity of $F$. For $C^1$ data, convexity gives, for any $a < \sigma_j(z)$,

$$U_{j,t}(z, \sigma_j(z)) \geq \frac{c - U_j(z,a)}{\sigma_j(z) - a} > 0. \tag{3.9}$$

The implicit function theorem gives $\sigma_j \in C^r$ whenever $F \in C^r$, for $1 \leq r \leq \infty$. Its real analytic version gives the same conclusion for $C^\omega$.

For $C^{0,\alpha}_{\text{loc}}$ data, omit the index $j$ and choose

$$a < b < \sigma(z_0) < d, \qquad U(z_0, a) < U(z_0, b).$$

The limits (3.5) permit this choice. By continuity, there are a relatively compact base ball $B$ around $z_0$ and $m > 0$ such that

$$b < \sigma(z) < d, \qquad \frac{U(z,b) - U(z,a)}{b-a} \geq m \qquad (z \in \overline{B}).$$

The local Hölder regularity of $U$ on a neighborhood of $\overline{B} \times [b,d]$ gives one constant $C$ such that

$$|U(z,t) - U(w,t)| \leq C|z-w|^\alpha \qquad (z, w \in B,\ t \in [b,d]).$$

Ordered secant slopes then yield

$$U(z,t) - U(z,s) \geq m(t-s), \qquad b \leq s \leq t \leq d.$$

If $\sigma(z) \geq \sigma(w)$, this implies

$$\begin{aligned} m|\sigma(z) - \sigma(w)| &\leq U(z, \sigma(z)) - U(z, \sigma(w)) \\ &= U(w, \sigma(w)) - U(z, \sigma(w)) \leq C|z-w|^\alpha. \end{aligned}$$

Interchanging $z, w$ gives the same bound in the other case. Hence

$$|\sigma(z) - \sigma(w)| \leq (C/m)|z-w|^\alpha, \qquad 0 < \alpha \leq 1. \tag{3.10}$$

Finally, suppose $U \in C^{k,\alpha}_{\text{loc}}$ with $k \geq 1$. The implicit function theorem first gives $\sigma \in C^k$. On a smaller convex coordinate neighborhood, $x \mapsto (x, \sigma(x))$ is Lipschitz and $U_t(x, \sigma(x))$ is bounded below by a positive constant. Differentiating the level equation gives

$$\sigma_{x_j} = -\frac{U_{x_j}(x, \sigma(x))}{U_t(x, \sigma(x))}, \qquad U_t(x, \sigma(x)) D^\beta \sigma = -P_\beta, \quad |\beta| = k,$$

where $P_\beta$ is a finite sum of products of derivatives of $U$ of order at most $k$, evaluated on the graph, and derivatives of $\sigma$ of order at most $k-1$. For $k = 1$, the first formula has $\alpha$-Hölder numerator and denominator. For $k \geq 2$,

$$D^j\sigma \in C^{0,1}_{\text{loc}}\ (j < k), \qquad D^j U \in C^{0,1}_{\text{loc}}\ (j < k), \qquad D^k U \in C^{0,\alpha}_{\text{loc}}.$$

Composition with the Lipschitz graph and multiplication preserve these local Hölder estimates. Division by $U_t$ therefore gives $D^\beta\sigma \in C^{0,\alpha}_{\text{loc}}$. Formula (3.8) now proves the assertion in every class. □

**Theorem 3.4** (Regularity criterion)**.** *Let $L \to X$ be a holomorphic line bundle over a compact complex manifold. For each class $\mathcal{R}$ in (3.1), the following are equivalent.*

(i) $\mathrm{Tot}(L^{-1})$ *admits a psh exhaustion of class* $\mathcal{R}$.
(ii) $L$ *admits a semipositive Hermitian metric of class* $\mathcal{R}$.

*Proof.* Proposition 3.3 gives (i)⇒(ii). Lemma 3.1 gives (ii)⇒(i). □

For $C^2$ data, the relation between the two Levi forms is explicit. Differentiating $U(z, \sigma(z)) = c$ twice and using $V(z, \zeta) = U(z, \mathrm{Re}\,\zeta)$ gives

$$\mathcal{L}_{-\sigma}(\xi) = \frac{\mathcal{L}_V(\xi, 2\partial\sigma(\xi))}{U_t} \geq 0. \tag{3.11}$$

The identity is evaluated along the graph $t = \sigma(z)$. For the dual norm, direct differentiation and completion of the square give

$$\mathcal{L}_{|\lambda|^2 e^{\varphi}}(\xi, \eta) = e^{\varphi}\left(|\eta + \lambda\partial\varphi(\xi)|^2 + |\lambda|^2 \mathcal{L}_{\varphi}(\xi)\right) \geq 0.$$

*Proof of Theorem 1.1.* Take the ruled surface $S$, the section $Y$, and the line bundle $L$ constructed in Section 2, for example with $F = K_B$. The total space $M = \mathrm{Tot}(L^{-1})$ has the compactification

$$M = \mathbf{P}(\mathcal{O}_S \oplus L^{-1}) \setminus \mathbf{P}(L^{-1}), \qquad v \longmapsto [(1, v)]. \tag{3.12}$$

Since $S$ is projective and $L$ is algebraic, (3.12) identifies $M$ as a smooth quasi-projective Kähler threefold.

Let $h$ be the locally Lipschitz semipositive metric of Proposition 2.6. Lemma 3.1 gives the locally Lipschitz psh exhaustion

$$\rho(v) = \|v\|^2_{h^{-1}}. \tag{3.13}$$

If $M$ admitted a $C^1$ psh exhaustion, Theorem 3.4 would give a $C^1$ semipositive Hermitian metric on $L$, contrary to Proposition 2.3. □

## 4. A non-Kähler modification over one point

The proof of Theorem 1.2 has two independent ingredients. First, modifications that are biholomorphic outside a compact set preserve the existence of psh exhaustions in each of the $C^r$ and local Hölder classes considered below. Second, Hironaka's construction produces a non-Kähler modification of any smooth threefold over a single point. We realize the latter by choosing the blowup centers in the exceptional plane of the blowup of that point.

**Proposition 4.1.** *Let $f : X' \to X$ be a proper surjective holomorphic map of complex manifolds, and suppose that*

$$f : X' \setminus f^{-1}(K) \longrightarrow X \setminus K$$

*is a biholomorphism for some compact $K \subset X$. For each class*

$$\mathcal{R} \in \{C^r : r \in \mathbb{Z}_{\geq 0} \cup \{\infty\}\} \cup \{C^{k,\alpha}_{\mathrm{loc}} : k \in \mathbb{Z}_{\geq 0},\ 0 < \alpha \leq 1\},$$

*$X'$ admits a psh exhaustion of class $\mathcal{R}$ if and only if $X$ does.*

*Proof.* If $u$ is a psh exhaustion of class $\mathcal{R}$ on $X$, then $u \circ f$ is psh and has the same regularity. Properness gives

$$\{u \circ f \leq a\} = f^{-1}(\{u \leq a\}) \Subset X'.$$

Conversely, let $F$ be a psh exhaustion of class $\mathcal{R}$ on $X'$. Choose a relatively compact open neighborhood $U$ of $K$. Since $f^{-1}(\overline{U})$ is compact, we may choose $c > \max_{f^{-1}(\overline{U})} F$. Let

$\chi : \mathbb{R} \to \mathbb{R}$ be smooth, convex, and nondecreasing, with $\chi = 0$ on $(-\infty, c]$ and $\chi(t) \to +\infty$ as $t \to +\infty$. Write $j : X \setminus K \to X' \setminus f^{-1}(K)$ for the holomorphic inverse of $f$, and define

$$G = \chi \circ F \circ j \quad \text{on } X \setminus K, \qquad G = 0 \quad \text{on } K.$$

The function $G$ vanishes throughout $U$. On $X \setminus K$, holomorphic pullback and smooth convex nondecreasing composition preserve plurisubharmonicity and the class $\mathcal{R}$. Thus $G$ is psh of class $\mathcal{R}$ on all of $X$. Finally, $G \circ f = \chi \circ F$, so surjectivity gives

$$\{G \le a\} = f(\{\chi \circ F \le a\}).$$

The right-hand side is compact because $\chi \circ F$ is an exhaustion. □

A non-constant real analytic function cannot be constant on a half-line while via our choice $\chi = 0$ on $(-\infty, c]$. Thus the real analytic class $C^\omega$ is excluded in our proof.

**Proposition 4.2.** *For every point $x$ of a connected smooth complex threefold $X$, there is a connected smooth non-Kähler threefold $N$ and a proper modification $\mu : N \to X$ that is biholomorphic over $X \setminus \{x\}$.*

*Proof.* We use Hironaka's opposite-order blowup construction, following [Bar15, §6]. First blow up $x$:

$$b : W = \mathrm{Bl}_x X \longrightarrow X, \qquad E = b^{-1}(x) \simeq \mathbf{P}^2.$$

In homogeneous coordinates $[u : v : w]$ on $E$, let

$$A = \{w = 0\}, \qquad B = \{uv = w^2\}.$$

These are smooth rational curves meeting transversely in $E$ at $p = [1 : 0 : 0]$ and $q = [0 : 1 : 0]$. Set

$$W_p = W \setminus \{q\}, \qquad W_q = W \setminus \{p\}, \qquad V = W_p \cap W_q = W \setminus \{p, q\}.$$

Form the two successive blowups

$$N_p = \mathrm{Bl}_{\widetilde{B \cap W_p}}(\mathrm{Bl}_{A \cap W_p} W_p),$$
$$N_q = \mathrm{Bl}_{\widetilde{A \cap W_q}}(\mathrm{Bl}_{B \cap W_q} W_q),$$

where a tilde denotes strict transform, and write $\nu_p : N_p \to W_p$ and $\nu_q : N_q \to W_q$ for the composite maps.

We check that the second centers $\widetilde{B \cap W_p}, \widetilde{A \cap W_q}$ are smooth. Since $A$ and $B$ meet transversely in the smooth surface $E$, there are coordinates $(z_1, z_2, z_3)$ centered at $p$ such that

$$E = \{z_3 = 0\}, \qquad A = \{z_2 = z_3 = 0\}, \qquad B = \{z_1 = z_3 = 0\}.$$

The first blowup has the chart

$$(z_1, z_2, t) \longmapsto (z_1, z_2, z_3 = z_2 t).$$

In this chart the strict transform of $B$ is the smooth curve $\{z_1 = t = 0\}$. Away from $p$, the first blowup is an isomorphism near $B \cap W_p$. The same argument at $q$, with $A$ and $B$ interchanged, treats the other order. Thus all blow-up centers are smooth, hence $N_p, N_q$ are smooth complex threefolds and $\nu_p, \nu_q$ are proper surjective holomorphic maps.

We now specify the gluing. Put

$$A_0 = A \cap V, \qquad B_0 = B \cap V, \qquad O_p = \nu_p^{-1}(V), \qquad O_q = \nu_q^{-1}(V).$$

The closed submanifolds $A_0, B_0$ of $V$ are disjoint. Blowups commute with restriction to open subsets. Over $V \setminus B_0$, both constructions therefore identify canonically with $\mathrm{Bl}_{A_0}(V \setminus B_0)$; over

$V\setminus A_0$, both identify canonically with $\mathrm{Bl}_{B_0}(V\setminus A_0)$. These identifications agree over $V\setminus(A_0\cup B_0)$, where they are the identity over the base. They consequently glue to a biholomorphism

$$\phi : O_p \xrightarrow{\sim} O_q, \qquad \nu_q \circ \phi = \nu_p.$$

Define the quotient

$$N = (N_p \sqcup N_q)/\!\sim, \qquad z \sim \phi(z) \quad (z \in O_p).$$

The canonical maps $j_p : N_p \to N$ and $j_q : N_q \to N$ are injective open embeddings. Indeed, for an open subset $G \subset N_p$, its saturation in $N_p \sqcup N_q$ is $G \sqcup \phi(G \cap O_p)$, which is open; the argument for $N_q$ is identical. Since the transition map is biholomorphic, the coordinate charts on the two pieces give a compatible complex atlas on $N$. The maps $\nu_p, \nu_q$ descend to a continuous map $\nu : N \to W$ satisfying

$$\nu^{-1}(W_p) = j_p(N_p), \qquad \nu^{-1}(W_q) = j_q(N_q).$$

For example, a point represented in $N_q$ whose image lies in $W_p$ lies over $V$, and hence also has a representative in $N_p$.

To verify Hausdorffness, consider two distinct points of $N$. If their images in $W$ are distinct, inverse images of disjoint open neighborhoods in $W$ separate them. If their images coincide at $w$, choose $W_k$ containing $w$. Both points then lie in the open Hausdorff subspace $j_k(N_k)$, where they have disjoint open neighborhoods. Thus $N$ is Hausdorff. It is second countable because it is covered by two second-countable open subspaces. The inherited atlas therewithal makes $N$ a smooth complex threefold, and $\nu$ is holomorphic.

We also verify properness. Given a compact subset $K \subset W$, choose finitely many relatively compact open subsets $U_\ell$ covering $K$, with each $\overline{U_\ell}$ contained in some $W_{k(\ell)}$. The compact sets $K_\ell = K \cap \overline{U_\ell}$ cover $K$, and

$$\nu^{-1}(K) = \bigcup_\ell j_{k(\ell)}(\nu_{k(\ell)}^{-1}(K_\ell))$$

is compact by properness of the local maps. Hence $\nu$ is proper. A blowup along a smooth center is proper and surjective with connected fibers, and therefore preserves connectedness. Thus $W$ is connected. Removing a point from a connected manifold of real dimension six preserves connectedness, so $W_p, W_q$, and then $N_p, N_q$, are connected. Their overlap is nonempty, so $N$ is connected. All blowup centers map into $E$, and consequently

$$\mu = b \circ \nu : N \longrightarrow X$$

is a proper surjective holomorphic map that is biholomorphic over $X \setminus \{x\}$. In particular, $\mu$ is a proper modification.

It remains to prove that $N$ is non-Kähler. We use the exceptional-curve relation underlying Hironaka's construction; compare [Sta, Tag 08J1]. On $N_p$, take the strict transform of the first exceptional divisor, and on $N_q$, take the second exceptional divisor. These divisors agree under $\phi$ and glue to a closed analytic divisor $D_A \subset N$ dominating $A$. Interchanging the roles of the two centers gives a closed analytic divisor $D_B$ dominating $B$. Equivalently,

$$D_A = \overline{\nu^{-1}(A \setminus \{p,q\})}, \qquad D_B = \overline{\nu^{-1}(B \setminus \{p,q\})}.$$

They carry induced maps

$$f_A : D_A \longrightarrow A, \qquad f_B : D_B \longrightarrow B.$$

Both divisors are compact, since they are closed subsets of the compact sets $\nu^{-1}(A)$ and $\nu^{-1}(B)$, respectively.

Let $S$ be the exceptional divisor of $\mathrm{Bl}_{A\cap W_p} W_p \to W_p$. In the first blowup chart at $p$ described above, it is given by $z_2 = 0$. Let $F_p \simeq \mathbf{P}^1$ be its complete fiber over $p$. The strict transform

$\widetilde{B} = \{z_1 = t = 0\}$ meets $F_p$ only at the normal direction $[z_2 : z_3] = [1 : 0]$, namely the point $c = (z_1, z_2, t) = (0, 0, 0)$, and meets $S$ transversely there. The second blowup is along the ideal $(z_1, t)$, leaving the coordinate $z_2$ unchanged. Its restriction to the strict transform of $S$ is therefore the surface blowup $\mathrm{Bl}_c S$. Thus the complete fiber of $f_A$ over $p$ has exactly two irreducible components: the strict transform $R_p$ of $F_p$ and the exceptional curve $T_p$ of this surface blowup. Both are smooth rational curves.

To determine their multiplicities, use the second blowup chart

$$(r, z_2, t) \longmapsto (z_1 = rt, z_2, t).$$

In this chart,

$$D_A = \{z_2 = 0\}, \qquad D_B = \{t = 0\},$$

and the indicated affine parts of the two curves are

$$R_p = \{r = z_2 = 0\}, \qquad T_p = \{t = z_2 = 0\}.$$

The map $f_A$ is $(r, t) \mapsto z_1 = rt$. Hence its fiber divisor is

$$f_A^*(p) = R_p + T_p,$$

with both multiplicities equal to one; the two components meet transversely at $r = t = 0$ on $D_A$.

The same complete curve $T_p$ is the fiber of $f_B$ over $p$. Indeed, the second exceptional divisor is locally a $\mathbf{P}^1$-bundle over $\widetilde{B}$, with projective fiber coordinate $[z_1 : t]$. Its fiber over $c$ is exactly the exceptional $\mathbf{P}^1$ in the slice $z_2 = 0$, that is, the exceptional curve of $\mathrm{Bl}_c S$. In the displayed chart, the map $f_B$ is $(r, z_2) \mapsto z_2$, confirming that its fiber is reduced. Consequently,

$$f_B^*(p) = T_p.$$

At $q$ the blowup order is reversed. The same calculation with $A$ and $B$ interchanged gives smooth rational curves $R_q, T_q$ such that

$$f_B^*(q) = R_q + T_q, \qquad f_A^*(q) = T_q.$$

These descriptions also show that $D_A, D_B$ are smooth: near each crossing one is a ruled surface and the other is a ruled surface blown up at a point; away from the crossings they are ruled surfaces.

Let $\alpha, \beta \in H_2(N, \mathbb{R})$ be the classes of general fibers of $f_A, f_B$, respectively. Since $A, B \simeq \mathbf{P}^1$, the fiber divisors of each map are linearly equivalent and hence homologous. Their images in $N$ give

$$\alpha = [R_p] + [T_p], \qquad \beta = [T_p], \qquad \beta = [R_q] + [T_q], \qquad \alpha = [T_q].$$

Thus $[R_p] = \alpha - \beta$ and $[R_q] = \beta - \alpha$, so

$$[R_p] + [R_q] = 0 \quad \text{in } H_2(N, \mathbb{R}). \tag{4.1}$$

If $N$ admitted a Kähler form $\omega$, its closedness would give

$$0 = \int_{R_p} \omega + \int_{R_q} \omega.$$

Both integrals are strictly positive because $R_p, R_q$ are nonempty compact holomorphic curves with their complex orientations. This contradiction proves that $N$ is non-Kähler. □

*Proof of Theorem 1.2.* We apply Proposition 4.2 to the threefold $M$ of Theorem 1.1 at $x$, and obtain a proper modification $\mu : N \to M$ with the center $\{x\}$. Hence by Proposition 4.1 with $K = \{x\}$, $\mu$ preserves the existence of a psh exhaustion both in $C^{0,1}_{\mathrm{loc}}$ and in $C^1$. Thus $N$ admits a locally Lipschitz psh exhaustion and admits no $C^1$ psh exhaustion, as required. □

## 5. Surface regularization and boundary criteria

Throughout this section, $X = S \setminus \operatorname{supp} D$, where $S$ is a smooth compact connected complex surface and $D \neq 0$ is effective. We identify reduced divisors with their supports. A *divergent psh collar* along a boundary component $E$ is a psh function on $U \setminus E$, for a neighborhood $U$ of $E$, tending uniformly to $+\infty$ as one approaches $E$. We prove regularization by constructing smooth collars and extending them to an exhaustion of $X$.

5.1. **Boundary criteria.** The following lemma about boundary criteria is needed for regularization. We utilize the intersection matrix of a connected boundary component to separate three cases: Negative definiteness obstructs an exhaustion, a positive eigenvalue gives a strictly psh collar, and a zero largest eigenvalue gives a divisor of square zero.

**Lemma 5.1.** *$X, S, D$ as per the above notations,*

(i) *If a connected component of* $\operatorname{supp} D$ *has negative definite intersection matrix, then $X$ has no continuous psh exhaustion.*

(ii) *Smooth divergent psh collars at all connected boundary components induce a smooth psh exhaustion of $X$.*

(iii) *For a connected reduced divisor $E = \bigcup E_j$ with simple normal crossings (SNC), let $\lambda$ be the largest eigenvalue of $I = (E_j \cdot E_k)$. If $\lambda > 0$, then $E$ has a smooth strictly psh divergent collar. If $\lambda = 0$, then $I \leq 0$,* $\dim \ker I = 1$, *and*

$$\exists\ A = \sum_j a_j E_j, \qquad a_j \in \mathbb{Z}_{>0}, \qquad A \cdot E_j = A^2 = 0. \tag{5.1}$$

*If moreover $S$ is projective, then $A$ is nef.*

*Proof.* For (i), let $E$ be a connected component of $\operatorname{supp} D$ with negative definite intersection matrix. By Grauert's contraction theorem [Gra62, §4, subsection 8(e)], there are a neighborhood $W$ of $E$, a normal complex surface $Z$, a point $z_0 \in Z$, and a proper surjective holomorphic map

$$\sigma : W \longrightarrow Z$$

such that

$$W \cap \operatorname{supp} D = E, \qquad \sigma^{-1}(z_0) = E,$$

and

$$\sigma|_{W \setminus E} : W \setminus E \xrightarrow{\sim} Z \setminus \{z_0\}$$

is a biholomorphism.

Suppose that $\rho$ is a continuous psh exhaustion of $X$. Then

$$u = \rho \circ \left(\sigma|_{W \setminus E}\right)^{-1}$$

is continuous and psh on $Z \setminus \{z_0\}$, and

$$u(y) \longrightarrow +\infty \qquad \text{as } y \longrightarrow z_0.$$

Indeed, a sequence $y_\nu \to z_0$ on which $u$ is bounded above would lift, by properness of $\sigma$, to a sequence accumulating on $E$. This contradicts compactness in $X$ of the corresponding sublevel set of $\rho$.

Choose a neighborhood $V$ of $z_0$ and a finite surjective holomorphic projection

$$\pi : V \longrightarrow \Delta^2, \qquad \pi^{-1}(0) = \{z_0\}.$$

For $\zeta \in \Delta^2 \setminus \{0\}$, define

$$v(\zeta) = \max_{y \in \pi^{-1}(\zeta)} u(y).$$

The map $\pi$ is proper and open, so $v$ is continuous on $\Delta^2 \setminus \{0\}$: properness gives upper semicontinuity, while openness gives lower semicontinuity. Here openness follows from the open mapping theorem for finite holomorphic maps from a pure two-dimensional complex space to a complex manifold of the same dimension.

Away from the analytic branch locus of $\pi$, the function $v$ is locally a finite maximum of psh functions, and is therefore psh. At branch values different from 0, continuity gives local boundedness above. The removable-singularity theorem for psh functions across analytic subsets then gives a psh extension, which agrees with $v$ by continuity. Thus $v$ is psh on $\Delta^2 \setminus \{0\}$.

Fix $0 < r < 1$ and put

$$M_r = \max_{\{|\zeta_1| \le r,\ |\zeta_2| = r\}} v < +\infty.$$

For each $a$ with $0 < |a| < r$, the function $\zeta_2 \mapsto v(a, \zeta_2)$ is subharmonic on the unit disc. The maximum principle therefore gives

$$v(a, \zeta_2) \le M_r \qquad (|\zeta_2| < r).$$

By continuity, the same bound holds when $\zeta_1 = 0$ and $0 < |\zeta_2| < r$. Writing $\Delta_r = \{\zeta \in \mathbb{C} : |\zeta| < r\}$, we obtain

$$u(y) \le v(\pi(y)) \le M_r \qquad \big(y \in \pi^{-1}(\Delta_r^2) \setminus \{z_0\}\big).$$

Since $\pi^{-1}(\Delta_r^2)$ is a neighborhood of $z_0$, this contradicts the divergence of $u$ at $z_0$ and proves (i).

For (ii), let $E_1, \dots, E_m$ be the connected components of $\operatorname{supp} D$. For each $j$, let $\Phi_j$ be a smooth divergent psh collar on $U_j \setminus E_j$, where $U_j$ is a neighborhood of $E_j$. Choose open neighborhoods $W_j$ of $E_j$ with pairwise disjoint closures such that

$$\overline{W_j} \subset U_j, \qquad W_j \cap \operatorname{supp} D = E_j.$$

Then $\partial W_j$ is a compact subset of $U_j \setminus E_j$. Choose

$$T_j > \max_{\partial W_j} \Phi_j$$

and a smooth convex nondecreasing function $\chi_j : \mathbb{R} \to [0, +\infty)$ such that $\chi_j = 0$ on $(-\infty, T_j]$ and $\chi_j(t) \to +\infty$ as $t \to +\infty$. Define a function on $X$ by

$$\rho_j(x) = \begin{cases} \chi_j(\Phi_j(x)), & x \in W_j \setminus E_j, \\ 0, & x \in X \setminus W_j. \end{cases}$$

By compactness of $\partial W_j$ and continuity of $\Phi_j$, the inequality $\Phi_j < T_j$ holds on a neighborhood of $\partial W_j$. Thus the two formulas agree there, and $\rho_j$ is smooth on $X$. On $W_j \setminus E_j$,

$$\mathcal{L}_{\rho_j}(\xi) = \chi_j'(\Phi_j)\mathcal{L}_{\Phi_j}(\xi) + \chi_j''(\Phi_j)|\partial\Phi_j(\xi)|^2 \ge 0.$$

Since $\rho_j$ vanishes near $\partial W_j$ and outside $W_j$, it is psh on all of $X$.

The finite sum

$$\rho = \sum_{j=1}^{m} \rho_j$$

is therefore a smooth nonnegative psh function on $X$. To prove that it is an exhaustion, fix $c \in \mathbb{R}$. The uniform divergence of $\Phi_j$ along $E_j$, together with $\chi_j(t) \to +\infty$, gives a neighborhood $V_{j,c} \subset W_j$ of $E_j$ such that $\rho_j > c$ on $V_{j,c} \setminus E_j$. Since all summands are nonnegative,

$$\{\rho \le c\} \subset S \setminus \bigcup_{j=1}^{m} V_{j,c}.$$

The right-hand side is a compact subset of $X$. The sublevel is closed in $X$ by continuity, and hence is compact. This proves (ii).

For (iii), write $E = \bigcup_{j=1}^{n} E_j$ and $I = (E_j \cdot E_k)_{j,k=1}^{n}$. Write $I = (I_{jk})_{1 \le j,k \le n}$, where $I_{jk} = E_j \cdot E_k$. For $j \neq k$, the SNC assumption implies that $E_j$ and $E_k$ meet transversely, with local intersection multiplicity one at each intersection point. Consequently,

$$I_{jk} = \#(E_j \cap E_k) \ge 0, \qquad I_{jk} > 0 \Longleftrightarrow E_j \cap E_k \neq \varnothing.$$

Consider the graph with vertices $1, \dots, n$, in which distinct vertices $j$ and $k$ are joined precisely when $I_{jk} > 0$. This graph is connected. Indeed, otherwise there would be a partition

$$\{1, \dots, n\} = J \sqcup K$$

into nonempty sets with $E_j \cap E_k = \varnothing$ whenever $j \in J$ and $k \in K$. The sets $\bigcup_{j \in J} E_j$ and $\bigcup_{k \in K} E_k$ would then be disjoint nonempty closed subsets whose union is $E$, contrary to the connectedness of $E$.

Choose

$$c > \max\{0, -I_{11}, \dots, -I_{nn}\}, \qquad B := I + c\,\mathrm{Id}_n.$$

Then $B$ is entrywise nonnegative and $B_{jj} > 0$ for every $j$. To verify irreducibility, let $j \neq k$ and choose a path

$$j = j_0, j_1, \dots, j_\ell = k$$

in the graph. Since every entry of $B$ is nonnegative, matrix multiplication gives

$$(B^\ell)_{jk} \ge B_{j_0 j_1} B_{j_1 j_2} \cdots B_{j_{\ell-1} j_\ell} > 0.$$

For $j = k$, we already have $B_{jj} > 0$. Thus, for every pair $j, k$, some positive power of $B$ has a strictly positive $(j,k)$-entry. This is the irreducibility criterion for a nonnegative matrix.

Let $r$ be the spectral radius of $B$. By the Perron–Frobenius theorem, $r$ is an algebraically simple eigenvalue of $B$, and there exists $v = (v_1, \dots, v_n)^T$ such that

$$v_j > 0 \quad (1 \le j \le n), \qquad Bv = rv.$$

Hence

$$Iv = (r - c)v.$$

Since $I$ is real symmetric, all its eigenvalues are real. If $\mu$ is any eigenvalue of $I$, then $\mu + c$ is an eigenvalue of $B$, so

$$\mu + c \le |\mu + c| \le r.$$

Therefore $\mu \le r - c$, proving that the largest eigenvalue of $I$ is

$$\lambda = r - c.$$

Adding a scalar multiple of the identity preserves eigenvalue multiplicities. Thus $\lambda$ is simple, and the vector $v$ above is a strictly positive eigenvector satisfying $Iv = \lambda v$.

Suppose first that $\lambda > 0$. Let $v = (v_1, \dots, v_n)^T$ be the positive eigenvector chosen above, so that

$$v_j > 0, \qquad (Iv)_j = \lambda v_j > 0 \quad (1 \le j \le n).$$

Set

$$\eta := \min_{1 \le j \le n} v_j > 0, \qquad C := \max_{1 \le j \le n} \sum_{k=1}^{n} |I_{jk}|.$$

Since $\lambda > 0$ is an eigenvalue of $I$, we have $C > 0$. Choose

$$0 < \varepsilon < \min\left\{ \frac{\eta}{2}, \frac{\lambda \eta}{2C} \right\}.$$

By the density of $\mathbb{Q}^n$ in $\mathbb{R}^n$, there exists $w = (w_1, \dots, w_n)^T \in \mathbb{Q}^n$ such that

$$|w_k - v_k| < \varepsilon \quad (1 \le k \le n).$$

Then

$$w_j \geq v_j - |w_j - v_j| > \eta - \varepsilon > \frac{\eta}{2} > 0.$$

Moreover, for every $j$,

$$\begin{aligned}(Iw)_j &= \lambda v_j + \sum_{k=1}^{n} I_{jk}(w_k - v_k) \\ &\geq \lambda v_j - \sum_{k=1}^{n} |I_{jk}|\,|w_k - v_k| \\ &\geq \lambda\eta - C\varepsilon > \frac{\lambda\eta}{2} > 0.\end{aligned}$$

Choose a positive integer $d$ that clears the denominators of all the $w_j$, and set $a_j := dw_j \in \mathbb{Z}_{>0}$. For

$$A := \sum_{k=1}^{n} a_k E_k, \qquad L := \mathcal{O}_S(A),$$

the symmetry of the intersection pairing gives

$$\deg(L|_{E_j}) = A \cdot E_j = \sum_{k=1}^{n} a_k (E_k \cdot E_j) = d(Iw)_j > 0 \quad (1 \leq j \leq n).$$

By Riemann–Roch on the compact Riemann surface $E_j$, some positive tensor power of $L|_{E_j}$ admits a nonzero holomorphic section. This section has a nonempty zero divisor, since its degree is positive. The only positive-dimensional irreducible compact analytic subsets of $E$ are its components $E_j$. Thus Grauert's positivity criterion [Gra62, §3, unnumbered lemma on p. 347] applies to $L|_E$. The embedding argument in [Gra62, §3, proof of Satz 2, pp. 343–345] then shows that $L|_E$ is ample. Consequently, for some integer $m > 0$, there is a holomorphic closed embedding

$$\iota : E \hookrightarrow \mathbb{P}^N, \qquad (L|_E)^{\otimes m} \simeq \iota^* \mathcal{O}_{\mathbb{P}^N}(1).$$

Let $h_E$ be the Hermitian metric on $L|_E$ whose $m$-th tensor power is the pullback of the Fubini–Study metric.

We construct a smooth metric on $L$ near $E$ that restricts to $h_E$. Choose sections $s_0, \ldots, s_N$ defining $\iota$. In a local frame $e$ of $L$, write $s_\ell = f_\ell e^{\otimes m}$ on $E$ and lift the holomorphic functions $f_\ell$ locally to holomorphic functions $F_\ell$ on $S$. After shrinking the neighborhood, the $F_\ell$ have no common zero. The weight

$$\varphi = \frac{1}{m} \log \left( \sum_{\ell=0}^{N} |F_\ell|^2 \right), \qquad \|e\|^2 = e^{-\varphi},$$

defines a smooth local metric restricting to $h_E$. Its curvature is positive in the tangent direction at every smooth point of $E$.

At a node $p$ of $E$, the Zariski tangent space of $E$ equals $T_pS$, since a local equation of $E$ is $zw = 0$. The closed embedding $\iota$ induces an injective map on Zariski tangent spaces. Consequently, the lifted holomorphic map $[F_0 : \cdots : F_N]$ has injective differential at $p$. Its pullback of the Fubini–Study form, and hence the curvature of the displayed metric, is positive definite at $p$ and on a sufficiently small neighborhood of $p$.

Choose such neighborhoods of the finitely many nodes to be pairwise disjoint. Patch the local metrics by taking a partition of unity of their logarithmic ratios relative to a fixed smooth reference metric $h_{\mathrm{ref}}$ on $L$. More explicitly, if the local metrics are $h_\alpha = e^{-\psi_\alpha} h_{\mathrm{ref}}$, set

$$h_0 = e^{-\sum_\alpha \theta_\alpha \psi_\alpha} h_{\mathrm{ref}},$$

where $\{\theta_\alpha\}$ is a subordinate smooth partition of unity on a neighborhood of $E$. Choose this partition so that the selected metric at each node is retained unchanged on a smaller node neighborhood. All local metrics restrict to $h_E$, so $h_0|_E = h_E$. Therefore $\Theta_{h_0}(L)$ is positive in the tangent direction along the smooth locus of $E$, and is positive definite on the chosen smaller node neighborhoods.

We next correct the normal curvature away from the nodes. Choose a smooth Hermitian metric $k$ on $\mathcal{O}_S(E)$, let $s_E$ be the canonical section of $\mathcal{O}_S(E)$, and put

$$\eta = \|s_E\|_k^2.$$

At a smooth point of $E$, choose coordinates $(z, w)$ with $E = \{w = 0\}$ and write $\eta = |w|^2 e^{-\gamma}$. Then

$$\partial\bar{\partial}\eta|_E = e^{-\gamma}|_E \, dw \wedge d\bar{w}.$$

Thus $dd^c\eta$ is semipositive along the smooth locus of $E$, vanishes on its tangent directions, and is strictly positive in the normal direction. At a node, the corresponding local expression is $\eta = |zw|^2 e^{-\gamma}$, so $dd^c\eta = 0$ at the node. In particular, $dd^c\eta$ is semipositive at every point of $E$.

Let $K$ be the compact subset of $E$ outside the smaller node neighborhoods. In tangent and normal coordinates along $K$, the Levi matrices of a local weight of $h_0$ and of $\eta$ have the respective forms

$$H_0 = \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \delta \end{pmatrix}, \qquad H_\eta = \begin{pmatrix} 0 & 0 \\ 0 & \kappa \end{pmatrix}, \qquad \alpha > 0, \quad \kappa > 0.$$

Using finitely many relatively compact coordinate charts covering $K$, the positive entries $\alpha, \kappa$ have uniform positive lower bounds, while $\beta, \delta$ are bounded. Hence a single constant $C > 0$ makes

$$\alpha(\delta + C\kappa) > |\beta|^2$$

throughout $K$. The metric

$$h = e^{-C\eta} h_0$$

therefore has positive curvature on $K$. At the remaining points of $E$, the curvature of $h_0$ is already positive definite, and adding $Cdd^c\eta$ preserves positivity there. Thus $\Theta_h(L) > 0$ at every point of $E$. For this fixed $C$, continuity gives a neighborhood $U$ of $E$ on which $\Theta_h(L) > 0$.

Let $s_A$ be the canonical section of $L = \mathcal{O}_S(A)$. On $U \setminus E$, define

$$\Phi = -\log \|s_A\|_h^2.$$

This function is smooth and satisfies

$$dd^c\Phi = \Theta_h(L) > 0.$$

Since all $a_j$ are positive, $s_A$ vanishes precisely on $E$. The smoothness of $h$ and compactness of $E$ imply that $\|s_A\|_h^2 \to 0$ uniformly as one approaches $E$. Hence $\Phi \to +\infty$ uniformly along $E$, proving the assertion for $\lambda > 0$.

Suppose now that $\lambda = 0$. Since $I$ is real symmetric and $\lambda$ is its largest eigenvalue, all eigenvalues of $I$ are nonpositive. The spectral theorem therefore gives

$$x^T I x \le 0 \qquad \text{for every } x \in \mathbb{R}^n,$$

so $I$ is negative semidefinite.

By the preceding Perron–Frobenius argument, the eigenvalue $\lambda = 0$ is simple and admits an eigenvector $v = (v_1, \ldots, v_n)^T$ with $v_j > 0$ for every $j$. Consequently,

$$\dim_{\mathbb{R}} \ker I = 1, \qquad \ker I = \mathbb{R}v, \qquad \operatorname{rank}_{\mathbb{R}} I = n - 1.$$

We next show that this kernel contains a vector with strictly positive integer entries. Every entry of $I$ is an integer. Since the rank of a matrix is the largest order of a nonzero minor, its rank is unchanged when computed over $\mathbb{Q}$ instead of $\mathbb{R}$. Thus

$$\operatorname{rank}_{\mathbb{Q}} I = n - 1.$$

Gaussian elimination over $\mathbb{Q}$ therefore produces a nonzero vector $w \in \mathbb{Q}^n$ satisfying $Iw = 0$. Because $\ker I = \mathbb{R}v$, we have

$$w = tv \qquad \text{for some } t \in \mathbb{R} \setminus \{0\}.$$

Replacing $w$ by $-w$ if necessary, we may assume $t > 0$. It follows that every coordinate of $w$ is strictly positive.

Choose a positive integer $d$ clearing all denominators of the coordinates of $w$, and put

$$a := dw = (a_1, \dots, a_n)^T \in \mathbb{Z}^n_{>0}.$$

Linearity gives $Ia = dIw = 0$. Hence the effective divisor

$$A := \sum_{j=1}^{n} a_j E_j$$

has support exactly $E$ and satisfies

$$A \cdot E_j = \sum_{k=1}^{n} a_k (E_k \cdot E_j) = (Ia)_j = 0 \qquad (1 \le j \le n).$$

By bilinearity of the intersection pairing,

$$A^2 = A \cdot \left( \sum_{j=1}^{n} a_j E_j \right) = \sum_{j=1}^{n} a_j (A \cdot E_j) = 0.$$

Finally, suppose that $S$ is projective. Let $C \subset S$ be any irreducible curve. If $C = E_j$ for some $j$, then $A \cdot C = 0$. Otherwise, $C$ is distinct from every $E_j$, so positivity of local intersection multiplicities gives $E_j \cdot C \ge 0$ for every $j$. Therefore

$$A \cdot C = \sum_{j=1}^{n} a_j (E_j \cdot C) \ge 0.$$

Thus $A$ has nonnegative intersection with every irreducible curve on $S$, and hence $A$ is nef. □

*Remark* 5.2 (Holomorphic convexity of surfaces). Let $X$ be a connected noncompact smooth complex surface admitting a continuous psh exhaustion. If $\mathcal{O}(X) \neq \mathbb{C}$, then $X$ is holomorphically convex.

The holomorphic-convexity assertion follows already from the argument of [Ohs81, Proposition 1.4, p. 156], which also applies when the psh exhaustion is only continuous. The fiber dichotomy used there requires only continuity, as Ohsawa explicitly observes after the proof of Theorem 1.1 on pp. 155–156. In the remaining argument, one applies Richberg regularization to continuous strictly psh functions on compact sublevel shells and then takes regular values of the resulting smooth functions to obtain the required strongly pseudoconvex boundaries. Thus differentiability of the original exhaustion is not needed. Alternatively, we may apply [Vîj22a, Theorem 1.1] directly, since it allows a general psh exhaustion without additional regularity assumptions.

*Remark* 5.3 (Holomorphic convexity and weak completeness). Every finite-dimensional holomorphically convex reduced complex space $X$, assumed Hausdorff and countable at infinity, is weakly 1-complete. Here smoothness on a complex space is understood through local ambient embeddings.

Let $r : X \to Y$ be the Remmert reduction. Then $Y$ is Stein and $r$ is proper. By [Nar60, Theorem 5], there is a proper holomorphic map $F = (F_1, \dots, F_N) : Y \to \mathbb{C}^N$. The function

$$\rho = \|F \circ r\|^2 = \sum_{j=1}^{N} |F_j \circ r|^2$$

is smooth and psh, and is an exhaustion because $F \circ r$ is proper.

*Remark* 5.4 (Line bundles over compact curves). Let $A$ be a holomorphic line bundle over a compact connected Riemann surface $B$. The following conditions are equivalent:

(i) $\mathrm{Tot}(A^{-1})$ admits a continuous psh exhaustion;
(ii) $\deg A \geq 0$;
(iii) $\mathrm{Tot}(A^{-1})$ admits a smooth psh exhaustion.

For (i)⇒(ii), Theorem 3.4 gives a continuous semipositive Hermitian metric on $A$. Its curvature is a positive current representing $c_1(A)$, so its total mass is $\deg A \geq 0$.

Conversely, suppose that $d := \deg A \geq 0$. Choose a smooth positive $(1,1)$-form $\omega$ on $B$ with $\int_B \omega = 1$, and a smooth Hermitian metric $h_0$ on $A$, with curvature $\theta_0$. Since $\int_B \theta_0 = d$,

$$\int_B (d\omega - \theta_0) = 0.$$

The solvability criterion for the Poisson equation on the compact connected Riemann surface $B$ gives a smooth real-valued function $f$ satisfying

$$dd^c f = d\omega - \theta_0.$$

Thus the metric $h := e^{-f} h_0$ has curvature

$$\Theta_h(A) = \theta_0 + dd^c f = d\omega \geq 0.$$

By Lemma 3.1, the function

$$\rho(v) := \|v\|^2_{h^{-1}}, \qquad v \in \mathrm{Tot}(A^{-1}),$$

is a smooth psh exhaustion. This proves (ii)⇒(iii), while (iii)⇒(i) is immediate.

We also record how the existence of these exhaustions behaves under modifications of the pair $(S, D)$. Let $\pi : \widetilde{S} \to S$ be an embedded resolution of the boundary, obtained by finitely many point blowups whose centers lie over $\operatorname{supp} D$, and put

$$\widetilde{D} := (\pi^* D)_{\mathrm{red}}.$$

Thus $\widetilde{D}$ is an SNC divisor and $\operatorname{supp} \widetilde{D} = \pi^{-1}(\operatorname{supp} D)$; in particular, it includes every exceptional curve of $\pi$. Each blowup is biholomorphic away from its center, so $\pi$ restricts to a biholomorphism

$$\widetilde{S} \setminus \operatorname{supp} \widetilde{D} \xrightarrow{\simeq} S \setminus \operatorname{supp} D = X.$$

This identifies continuous, respectively smooth, psh exhaustions on the two complements. Hence, for the question of existence of such exhaustions, we may resolve the boundary and assume that it is SNC.

For a one-point blowup, let $b : S' \to S$ be the blowup of a point $x \in X$, and let $D'$ be the strict transform of $D$. Since $x \notin \operatorname{supp} D$, we have

$$\operatorname{supp} D' = b^{-1}(\operatorname{supp} D), \qquad X' := S' \setminus \operatorname{supp} D' = b^{-1}(X).$$

Thus the exceptional curve $F := b^{-1}(x) \simeq \mathbf{P}^1$ remains in $X'$, and the restriction $b|_{X'} : X' \to X$ is proper and surjective, with

$$b : X' \setminus F \xrightarrow{\simeq} X \setminus \{x\}.$$

Proposition 4.1, applied with $K = \{x\}$ and $r = 0$ or $r = \infty$, shows that $X'$ admits a continuous, respectively smooth, psh exhaustion if and only if $X$ does. Iterating gives the same conclusion for any finite sequence of one-point blowups.

5.2. **Current averaging and regularization.** The following theorem and its quasi-projective corollary together prove Theorem 1.3.

**Theorem 5.5.** *Let $S$ be a compact Kähler surface and $D = \sum_j D_j \neq 0$ a reduced divisor. Suppose $X = S \setminus D$ admits a continuous psh exhaustion. Then $X$ admits a smooth psh exhaustion. If $X$ is not a modification of a Stein space, there is a divisor*

$$A = \sum_j a_j D_j, \qquad a_j \in \mathbb{Z}_{>0}, \qquad A \cdot D_j = A^2 = 0,$$

*such that $\mathcal{O}_S(A)$ has a smooth Hermitian metric with zero curvature near $D$. The canonical section of $A$ induces a pluri-harmonic divergent collar along $D$.*

We now use the boundary criteria to prove the theorem and therewithal regularization for complements of divisors in compact Kähler surfaces. When the complement is not a modification of a Stein space, we use the given exhaustion to produce by an averaging process (Lemma 5.7) a positive $dd^c$-closed currents with compact support on $X$, and the identify it with a positive multiple of the class of a boundary divisor of square zero (Lemma 5.8). This yields a divergent pluri-harmonic collar, which, after gluing by Lemma 5.1(ii), extends to a smooth psh exhaustion.

*Remark* 5.6. A proper modification of a Stein space is holomorphically convex, but the converse need not hold.

**Lemma 5.7.** *Let $(S, \omega)$ be a compact Kähler surface and $X \subset S$ a connected noncompact open set with a continuous psh exhaustion $\rho$. If $X$ is not a modification of a Stein space, there is a positive $(1,1)$-current $T$ on $S$ such that*

$$dd^c T = 0, \qquad \int_S T \wedge \omega = 1, \qquad \operatorname{supp} T \Subset X, \qquad \nu(T, x) = 0 \quad (x \in S). \tag{5.2}$$

*Here $\nu(T, x)$ denotes the Lelong number of $T$ at $x$.*

*Proof.* Let $\mathcal{E}^0(X)$ denote the family of continuous real-valued psh exhaustions of $X$. For $u \in \mathcal{E}^0(X)$, the strictly psh locus of $u$ is given by

$$\operatorname{Str}(u) := \{x \in X : dd^c u \geq c\omega \text{ on some neighborhood of } x \text{ for some } c > 0\},$$

where the inequality is understood in the sense of currents. Then the locus where every continuous psh exhaustion fails to be strictly psh is given by

$$\Sigma_X^0 := \bigcap_{u \in \mathcal{E}^0(X)} (X \setminus \operatorname{Str}(u)).$$

[MS20, Lemma 2.1] states that if $\rho(\Sigma_X^0)$ contains no half-line $(t, \infty)$, then $X$ must be a modification of a Stein space. Therefore, here we can choose $t_0 \in \mathbb{R}$ such that

$$(t_0, \infty) \subset \rho(\Sigma_X^0). \tag{5.3}$$

For $t > t_0$, no smooth strictly psh function exists near $K_t = \{\rho = t\}$. Indeed, if $v$ were such a function, then there is a $\delta > 0$ such that $v$ is strictly psh on the compact shell $\{t - \delta \leq \rho \leq t + \delta\}$.

After scaling and adding a constant, we may assume $0 < v < 1$. For $0 < \eta < 1$, $0 < \varepsilon < \eta\delta/2$, set $g(s) = s + \eta|s - t|$. The function

$$\psi = \begin{cases} \max\{g(\rho), \rho + \varepsilon v\}, & |\rho - t| < \delta, \\ g(\rho), & |\rho - t| \geq \delta \end{cases}$$

is well defined since near $|\rho - t| = \delta$, $g(\rho) - \rho = \eta|\rho - t| > \varepsilon v$. Both entries of the maximum are psh, since $g$ is convex and increasing. On $K_t$,$\rho + \varepsilon v > g(\rho)$, so $\psi = \rho + \varepsilon v$ is strictly psh near $K_t$. Finally $\psi \geq g(\rho)$ makes $\psi$ an exhaustion. Thus $\psi \in \mathcal{E}^0(X)$. Since $v$ is smooth and strictly psh near the compact set $K_t$, there is $c > 0$ such that $dd^c v \geq c\omega$ on a neighborhood of $K_t$. On a smaller neighborhood, $\psi = \rho + \varepsilon v$, so

$$dd^c\psi = dd^c\rho + \varepsilon dd^c v \geq \varepsilon c\omega.$$

Hence $K_t \subset \mathrm{Str}(\psi)$, and the definition of $\Sigma_X^0$ gives $K_t \cap \Sigma_X^0 = \varnothing$. This contradicts (5.3), since $t > t_0$.

Let $\mathcal{B}_t$ be the Banach space of continuous real $(1,1)$-forms on $K_t$, with the uniform norm induced by $\omega$. Consider the subspace and open cone

$$V_t = \{dd^c f|_{K_t} : f \in C^\infty(S, \mathbb{R})\}, \qquad P_t = \{\eta \in \mathcal{B}_t : \eta > 0 \text{ on } K_t\}.$$

By the above, $V_t \cap P_t = \varnothing$. The cone $P_t$ is nonempty, open, and convex; hence we also have $\overline{V_t} \cap P_t = \varnothing$. By Hahn–Banach separation, there is a nonzero continuous real linear functional $\ell_t$ on $\mathcal{B}_t$ and $a \in \mathbb{R}$ such that

$$\ell_t(v) \leq a \leq \ell_t(p) \qquad (v \in V_t,\ p \in P_t).$$

Since $V_t$ is a linear subspace, the first inequality forces $\ell_t|_{V_t} = 0$, so $a \geq 0$. For fixed $p \in P_t$, applying the second inequality to $rp$ and letting $r \downarrow 0$ gives $a \leq 0$. Hence $a = 0$ and $\ell_t \geq 0$ on $P_t$.

Write $\omega_t = \omega|_{K_t}$. If $\eta \geq 0$, then $\eta + \delta\omega_t \in P_t$ for every $\delta > 0$; letting $\delta \downarrow 0$ gives $\ell_t(\eta) \geq 0$. Moreover, $\ell_t(\omega_t) > 0$. Indeed, compactness of $K_t$ gives, for each $\eta \in \mathcal{B}_t$, a constant $C > 0$ such that

$$-C\omega_t \leq \eta \leq C\omega_t,$$

and therewithal $-C\ell_t(\omega_t) \leq \ell_t(\eta) \leq C\ell_t(\omega_t)$. If $\ell_t(\omega_t) = 0$, then $\ell_t(\eta) = 0$ for every $\eta$, contradicting $\ell_t \neq 0$.

For smooth real $(1,1)$-forms $\eta$ on $S$, define

$$\langle T_t, \eta\rangle := \frac{\ell_t(\eta|_{K_t})}{\ell_t(\omega_t)},$$

and extend complex linearly. Continuity in the uniform norm makes $T_t$ a current of order zero. It is positive and supported on $K_t$, since its value depends only on $\eta|_{K_t}$. For every $f \in C^\infty(S, \mathbb{R})$,

$$\langle dd^c T_t, f\rangle = \langle T_t, dd^c f\rangle = 0,$$

because $dd^c f|_{K_t} \in V_t$. Therefore, we have

$$T_t \geq 0, \qquad dd^c T_t = 0, \qquad \operatorname{supp} T_t \subset K_t, \qquad \int_S T_t \wedge \omega = 1. \tag{5.4}$$

Fix $t_0 < a < b$ and define

$$t_{n,j} = a + \frac{j - \frac{1}{2}}{n}(b - a), \qquad T_n = \frac{1}{n}\sum_{j=1}^{n} T_{t_{n,j}}.$$

Each $T_n$ is positive, $dd^c$-closed, has mass one, and is supported in the compact set $K = \rho^{-1}([a,b]) \Subset X$. Weak compactness of positive currents with bounded mass gives a subsequence $T_{n_k} \rightharpoonup T$. Positivity and $dd^c$-closedness pass to the limit, as do the support condition and the mass normalization. Thus

$$T \geq 0, \qquad dd^c T = 0, \qquad \operatorname{supp} T \subset K, \qquad \int_S T \wedge \omega = 1.$$

We show that $T$ has zero Lelong numbers. Fix $x \in X$ and choose holomorphic coordinates $z$ centered at $x$. Write $B_r = \{|z| < r\}$, with $B_{2R} \Subset X$, and set $\beta = dd^c|z|^2$. Lelong–Jensen monotonicity for positive $dd^c$-closed currents [DNS22, Appendix B, (B.1)–(B.2)] gives

$$r^{-2} \int_{B_r} T_t \wedge \beta \leq R^{-2} \int_{B_R} T_t \wedge \beta \leq C_R \qquad (0 < r < R). \tag{5.5}$$

Here $C_R$ is independent of $t$: indeed, $\beta \leq c_R \omega$ on $B_R$ and $\int_S T_t \wedge \omega = 1$, so we may take $C_R = c_R/R^2$.

Put

$$m_r = \inf_{B_r} \rho, \qquad M_r = \sup_{B_r} \rho.$$

Since $\operatorname{supp} T_t \subset \{\rho = t\}$, the current $T_t$ vanishes on $B_r$ whenever $t \notin [m_r, M_r]$. The grid points $t_{n,j}$ are spaced by $(b-a)/n$; hence the number of potentially contributing terms satisfies

$$N_n(r) := \#\{j : t_{n,j} \in [m_r, M_r]\} \leq \frac{n(M_r - m_r)}{b-a} + 2.$$

Averaging (5.5) therefore yields

$$r^{-2} \int_{B_r} T_n \wedge \beta \leq \frac{C_R N_n(r)}{n} \leq C_R \left( \frac{M_r - m_r}{b-a} + \frac{2}{n} \right). \tag{5.6}$$

To pass this estimate to the limit, fix $0 < r < R$ and $\chi \in C_c^\infty(B_r)$ with $0 \leq \chi \leq 1$. Weak convergence and (5.6) give

$$\begin{aligned} \langle T, \chi\beta \rangle &= \lim_{k\to\infty} \langle T_{n_k}, \chi\beta \rangle \\ &\leq \limsup_{k\to\infty} C_R r^2 \left( \frac{M_r - m_r}{b-a} + \frac{2}{n_k} \right) \\ &= \frac{C_R r^2}{b-a} (M_r - m_r). \end{aligned}$$

Since $T \wedge \beta$ is a positive measure, taking the supremum over such $\chi$ gives

$$0 \leq r^{-2} \int_{B_r} T \wedge \beta \leq \frac{C_R}{b-a} (M_r - m_r). \tag{5.7}$$

Continuity of $\rho$ at $x$ implies $M_r - m_r \to 0$ as $r \downarrow 0$. Thus

$$\lim_{r\downarrow 0} r^{-2} \int_{B_r} T \wedge \beta = 0,$$

which gives $\nu(T, x) = 0$ by the local density formula for Lelong numbers. This holds for every $x \in X$. Finally, $\operatorname{supp} T \subset K \Subset X$ implies that $T$ vanishes near every point of $S \setminus X$. Hence $\nu(T, x) = 0$ for all $x \in S$. □

**Lemma 5.8.** *For a positive $dd^c$-closed $(1,1)$-current $T$ on a compact Kähler surface with all Lelong numbers zero, there is a class $\alpha_T \in H^{1,1}(S, \mathbb{R})$ satisfying*

$$\alpha_T \cdot [\eta] = \int_S T \wedge \eta \qquad (d\eta = 0), \tag{5.8}$$

*for every smooth real* $(1,1)$*-form* $\eta$*. Moreover,*

$$\alpha_T^2 \geq 0, \qquad dT \neq 0 \implies \alpha_T^2 > 0. \tag{5.9}$$

*If* $dT = 0$*, then* $\alpha_T = [T]$*.*

*Proof.* By [DNS22, Lemma B.2], there is a decomposition

$$T = \Omega + \partial R + \bar{\partial}\overline{R} + \sqrt{-1}\partial\bar{\partial}u$$

with $\Omega$ a smooth real closed $(1,1)$-form, $R$ a $(0,1)$-current, and $u$ a real distribution. Set $\alpha_T = [\Omega] \in H^{1,1}(S,\mathbb{R})$. If $\eta$ is a smooth real closed $(1,1)$-form, then $\partial\eta = \bar{\partial}\eta = 0$, since these two terms have distinct bidegrees. Integration by parts on $S$ gives

$$\langle T - \Omega, \eta\rangle = 0, \qquad \int_S T \wedge \eta = \int_S \Omega \wedge \eta = \alpha_T \cdot [\eta].$$

This proves (5.8). The intersection pairing on $H^{1,1}(S,\mathbb{R})$ is nondegenerate, so these pairings determine $\alpha_T$ uniquely; in particular, it is independent of the decomposition. If $dT = 0$, its usual cohomology class $[T] \in H^{1,1}(S,\mathbb{R})$ satisfies the same pairing identities. Uniqueness therefore gives $\alpha_T = [T]$.

The forms $\bar{\partial}R$ and $\partial\overline{R}$ belong to $L^2$ by the same decomposition. Hence the energy

$$E(T) := \int_S \bar{\partial}R \wedge \partial\overline{R}$$

is finite and nonnegative. It is independent of the decomposition, and $E(T) = 0$ if and only if $dT = 0$ [DNS22, Appendix B, (B.4) and the subsequent discussion].

Since all Lelong numbers vanish, the set $\{x \in S : \nu(T,x) > 0\}$ is empty. Thus the no-mass hypotheses of [DNS22, Theorem 2.2] hold for $T_1 = T_2 = T$. Choose a tangent current of $T \otimes T$ along the diagonal, and let $\vartheta$ be its associated positive measure on the diagonal. The mass formula [DNS22, Theorem 2.2(2), (2.3)] gives

$$\|\vartheta\| = \int_S \Omega \wedge \Omega - 2E(T), \qquad \alpha_T^2 = \|\vartheta\| + 2E(T) \geq 2E(T) \geq 0.$$

If $dT \neq 0$, then $E(T) > 0$, so $\alpha_T^2 > 0$. This proves (5.9). □

*Proof of Theorem 5.5.* If $r : X \to Y$ is a proper modification of a Stein space, choose a smooth strictly psh exhaustion $\tau$ of $Y$. Then $\tau \circ r$ is smooth and psh, and its sublevel sets are compact by properness of $r$. This proves the first assertion in this case.

Suppose henceforth that $X$ is not a modification of a Stein space, and fix a Kähler form $\omega$ on $S$. Lemma 5.7 gives a positive $dd^c$-closed current $T$ of mass one, with support in a compact set $K \Subset X$ and with all Lelong numbers zero. By Lemma 5.8, its associated class $\alpha = \alpha_T \in H^{1,1}(S,\mathbb{R})$ satisfies

$$\alpha \cdot [\omega] = 1, \qquad \alpha^2 \geq 0, \qquad dT \neq 0 \Longrightarrow \alpha^2 > 0. \tag{5.10}$$

In particular, $\alpha \neq 0$.

We first show that $\alpha$ is orthogonal to every boundary curve. For each $D_j$, choose a smooth Hermitian metric $k_j$ on $\mathcal{O}_S(D_j)$ with curvature $\theta_j$ representing $[D_j]$, and let $s_j$ be its canonical section. Choose $\chi \in C_c^\infty(X)$ equal to one near $K$. Since $s_j$ is nonvanishing on $X$, the function $v_j = \chi \log \|s_j\|_{k_j}^2$, extended by zero, is smooth on $S$. Poincaré–Lelong gives $dd^c v_j = -\theta_j$ near $K$. Using $\operatorname{supp} T \subset K$ and $dd^c T = 0$, we obtain

$$\alpha \cdot [D_j] = \langle T, \theta_j\rangle = -\langle T, dd^c v_j\rangle = -\langle dd^c T, v_j\rangle = 0. \tag{5.11}$$

Let $E$ be a connected component of $D$, and let $\lambda_E$ be the largest eigenvalue of its intersection matrix $I_E$. If $\lambda_E < 0$, then $I_E$ is negative definite, contradicting Lemma 5.1(i) and the existence of a continuous psh exhaustion of $X$. If $\lambda_E > 0$, choose a real eigenvector $b = (b_j) \neq 0$ and set

$$\beta = \sum_{D_j \subset E} b_j [D_j].$$

Then

$$\beta^2 = b^{\mathsf{T}} I_E b = \lambda_E \sum_j b_j^2 > 0, \qquad \alpha \cdot \beta = 0.$$

The Hodge index theorem makes the intersection form negative definite on $\beta^{\perp}$. Since $\alpha \neq 0$, this would give $\alpha^2 < 0$, contrary to (5.10). Therefore $\lambda_E = 0$.

The off-diagonal entries of $I_E$ are nonnegative, and its intersection graph is connected. For sufficiently large $q > 0$, the matrix $I_E + q\,\mathrm{Id}$ is nonnegative and irreducible. Perron–Frobenius shows that its largest eigenvalue $q$ is simple and has a strictly positive eigenvector. Thus $\ker I_E$ is one-dimensional and contains a strictly positive vector. Since $I_E$ has integer entries, its kernel has a rational generator; choosing its sign and clearing denominators gives positive integers $a_j$ such that

$$A_E := \sum_{D_j \subset E} a_j D_j, \qquad A_E \cdot D_j = 0 \quad (D_j \subset E).$$

Distinct connected boundary components are disjoint. Summing the divisors $A_E$ over these components therefore gives

$$A = \sum_j a_j D_j, \qquad a_j \in \mathbb{Z}_{>0}, \qquad \operatorname{supp} A = D, \qquad A \cdot D_j = A^2 = 0. \tag{5.12}$$

We next identify $\alpha$. Since $A$ is nonzero and effective, $\int_A \omega > 0$. Put

$$c = \left( \int_A \omega \right)^{-1}, \qquad v = \alpha - c[A].$$

By (5.11), we have $\alpha \cdot [A] = 0$, and hence

$$v \cdot [\omega] = 1 - c \int_A \omega = 0, \qquad v^2 = \alpha^2 - 2c\alpha \cdot [A] + c^2 A^2 = \alpha^2 \geq 0.$$

By the Hodge index theorem, the intersection form is negative definite on $[\omega]^{\perp}$, so $v = 0$. Consequently $\alpha = c[A]$ and $\alpha^2 = c^2 A^2 = 0$. By (5.10), this forces $dT = 0$; Lemma 5.8 then identifies $\alpha$ with $[T]$. Thus

$$dT = 0, \qquad [T] = \alpha = c[A], \qquad c = \left( \int_A \omega \right)^{-1} > 0. \tag{5.13}$$

The closed currents $T$ and $c[A]$ therefore represent the same cohomology class. The $dd^c$-lemma for currents on the compact Kähler surface $S$ supplies a real distribution $u$ such that

$$dd^c u = T - c[A]. \tag{5.14}$$

Here $[A]$ denotes the current of integration over $A$. Choose a neighborhood $U$ of $D$ disjoint from $K$. On a holomorphic trivializing chart $U_j \subset U$ for $\mathcal{O}_S(A)$, write its canonical section as $s_A = f_j e_j$. Since $T = 0$ on $U$, Poincaré–Lelong gives

$$dd^c (u + c \log |f_j|^2) = -c[A] + c[A] = 0 \qquad \text{on } U_j$$

in the sense of distributions. Taking the trace in local coordinates shows that this distribution is harmonic. Elliptic regularity therefore makes it smooth, and its vanishing $dd^c$ makes it pluriharmonic. Thus

$$u = -c\log|f_j|^2 + h_j, \qquad h_j \text{ smooth and pluriharmonic on } U_j. \tag{5.15}$$

Set $b_j = h_j/c$. On an overlap, write $e_k = g_{jk}e_j$. Then $f_k = g_{jk}^{-1}f_j$, so

$$b_k = b_j - \log|g_{jk}|^2, \qquad dd^c b_j = 0.$$

These identities hold first off $D$ and then everywhere on the overlap by smoothness. Consequently, $\|e_j\|^2_{h_{\mathrm{flat}}} = e^{-b_j}$ defines a smooth Hermitian metric on $\mathcal{O}_S(A)|_U$ with zero curvature.

To extend the metric globally, choose a smooth Hermitian metric $h_0$ on $\mathcal{O}_S(A)$. There is a smooth real function $b$ on $U$ such that $h_{\mathrm{flat}} = e^{-b}h_0$. Choose $\zeta \in C_c^\infty(U)$ equal to one on a smaller neighborhood $U'$ of $D$, and extend $\zeta b$ by zero to $S$. Then

$$h = e^{-\zeta b}h_0$$

is a global smooth Hermitian metric agreeing with $h_{\mathrm{flat}}$ on $U'$, and hence has zero curvature there.

On $U' \setminus D$, define

$$\Phi := -\log\|s_A\|_h^2 = -\log|f_j|^2 + b_j = u/c.$$

This is a smooth pluriharmonic function. Since every coefficient of $A$ is positive, the zero locus of $s_A$ is exactly $D$. The smoothness of $h$ and compactness of $D$ imply that $\|s_A\|_h^2$ tends uniformly to zero along $D$. Thus $\Phi$ tends uniformly to $+\infty$, giving a smooth divergent psh collar along every connected boundary component. Lemma 5.1(ii) extends these collars to a smooth psh exhaustion of $X$. □

**Corollary 5.9.** *Every smooth complex quasi-projective surface with a continuous psh exhaustion has a smooth psh exhaustion.*

*Proof.* Let $\rho$ be a continuous psh exhaustion of $X$. Since $X$ is quasi-projective, it has finitely many connected components, each open and closed. The restriction of $\rho$ to each component is again an exhaustion, and compact components admit constant smooth psh exhaustions. We first treat a connected noncompact component, still denoted by $X$.

Resolve a projective closure of $X$, preserving its smooth open subset $X$, and then principalize the boundary ideal by blowups supported outside $X$ [Wło05, Theorems 1.0.1 and 1.0.2]. The boundary may contain isolated points, which principalization also replaces by divisors. This gives a smooth projective surface $S$ and a reduced simple normal crossings divisor $D \neq 0$ with $X \simeq S \setminus D$. Theorem 5.5 now supplies a smooth psh exhaustion of $X$.

Applying this construction to every noncompact component and choosing constants on the compact components gives a smooth psh function on the original surface. Its sublevel sets are finite unions of compact sets, so it is an exhaustion. □

**Corollary 5.10.** *Let $C \subset S$ be a smooth connected square-zero curve in a compact Kähler surface. A continuous divergent psh collar at $C$ implies a smooth psh exhaustion of $S \setminus C$. If the complement is not a modification of a Stein space, $\mathcal{O}_S(C)$ is unitary flat near $C$, and $C$ has a pluriharmonic divergent collar.*

*Proof.* Let $\Phi$ be the collar, choose a neighborhood $W$ of $C$ such that $\Phi$ is defined near $\overline{W} \setminus C$, and take $M > \sup_{\partial W}\Phi$. Set

$$\rho(x) = \begin{cases} \max\{\Phi(x), M\}, & x \in W \setminus C, \\ M, & x \notin W. \end{cases}$$

The formulas agree near $\partial W$, and divergence at $C$ makes $\rho$ a continuous psh exhaustion. In Theorem 5.5, the single-component kernel generator can be chosen to be 1, so $A = C$. □

5.3. **Numerical existence criteria.** We next use the classification of compact surfaces to obtain a numerical criterion for nonprojective surfaces, followed by the corresponding criterion for smooth ruled surfaces.

**Theorem 5.11.** *Let $S$ be a smooth compact connected nonprojective surface and $D \neq 0$ an effective divisor. The following conditions are equivalent.*

(i) *$S \setminus \operatorname{supp} D$ admits a continuous psh exhaustion.*
(ii) *$S \setminus \operatorname{supp} D$ admits a smooth psh exhaustion.*
(iii) *no connected boundary component has negative definite intersection matrix.*

*Proof.* Put $X = S \setminus \operatorname{supp} D$. The implication (ii)⇒(i) is immediate, and Lemma 5.1(i) gives (i)⇒(iii). Assume henceforth (iii).

By the Chow–Kodaira projectivity theorem [CK52], in its general form [Kod60, Theorem 3.1], a compact complex surface of algebraic dimension two is projective. Thus $a(S) \in \{0, 1\}$. We treat these cases using [FN05, §2].

**Case $a(S) = 1$.** The algebraic reduction is a holomorphic elliptic fibration

$$f : S \longrightarrow B$$

onto a compact Riemann surface, with connected fibers, and every irreducible curve on $S$ is contained in a fiber.

Fix a fiber

$$F = f^*(p) = \sum_{j=1}^{r} m_j C_j, \qquad m_j > 0.$$

Since $\mathcal{O}_S(F) = f^*\mathcal{O}_B(p)$ restricts trivially to each $C_j$, we have $F \cdot C_j = 0$. Hence

$$m_j C_j^2 = -\sum_{k \neq j} m_k (C_j \cdot C_k).$$

For $I = (C_j \cdot C_k)$, substitution gives

$$x^{\mathsf{T}} I x = -\sum_{j<k} (C_j \cdot C_k) m_j m_k \left( \frac{x_j}{m_j} - \frac{x_k}{m_k} \right)^2. \tag{5.16}$$

The intersection graph of the fiber is connected. Consequently, equality holds precisely when $x_j/m_j$ is independent of $j$. In particular, every proper principal submatrix of $I$ is negative definite: extend a vector on the corresponding proper subset by zero, and observe that equality forces all its coordinates to vanish.

Every connected component of $\operatorname{supp} D$ lies in a fiber. If it were a proper subset of that fiber's irreducible components, its intersection matrix would be negative definite, contrary to (iii). Thus $\operatorname{supp} D$ is a union of complete fiber supports:

$$\operatorname{supp} D = f^{-1}(P), \qquad X = f^{-1}(B \setminus P),$$

where $P \subset B$ is finite and nonempty.

The open Riemann surface $B \setminus P$ admits a smooth subharmonic exhaustion $\tau$. Its pullback $\tau \circ f$ under the proper map $X \to B \setminus P$ is a smooth psh exhaustion of $X$. This proves (ii) in this case.

**Case $a(S) = 0$.** By [FN05, Lemma 2.1], every holomorphic line bundle $L$ on $S$ satisfies $c_1(L)^2 \leq 0$. For a connected boundary component $E$, this implies that its intersection matrix

$I_E$ is negative semidefinite: the inequality holds on rational vectors by clearing denominators and applying it to integral divisors, and then on real vectors by continuity. Condition (iii) therefore forces the largest eigenvalue of $I_E$ to be zero.

The off-diagonal entries of $I_E$ are nonnegative, and its intersection graph is connected. Applying Perron–Frobenius to $I_E + q\,\mathrm{Id}$ for $q > 0$ sufficiently large shows that $\ker I_E$ is one-dimensional and contains a strictly positive vector. Since $I_E$ has integer entries, its kernel has a rational generator. Choosing its sign and clearing denominators gives positive integral coefficients. Doing this on every connected boundary component yields an effective divisor

$$A = \sum_j a_j D_j, \qquad a_j \in \mathbb{Z}_{>0}, \qquad \operatorname{supp} A = \operatorname{supp} D, \qquad A \cdot D_j = A^2 = 0.$$

Let $p : S \to S_0$ be the contraction to the minimal model. We claim that $A = p^*A_0$ for a nonzero effective divisor $A_0$ on $S_0$ with $A_0^2 = 0$. To verify this at one step, let $q : T \to T'$ contract a $(-1)$-curve $E$, and let $G$ be the effective square-zero divisor obtained on $T$. Put $G' = q_*G$ and write

$$G = q^*G' + bE, \qquad b \in \mathbb{Z}.$$

Since algebraic dimension is bimeromorphically invariant, $a(T') = 0$, and hence $(G')^2 \le 0$ by [FN05, Lemma 2.1]. Using $q^*G' \cdot E = 0$ and $E^2 = -1$, we obtain

$$0 = G^2 = (G')^2 - b^2.$$

Thus $b = 0$ and $(G')^2 = 0$. Moreover, $G' \neq 0$, since otherwise $G = 0$. Iteration proves the claim.

In particular,

$$A = p^*A_0, \qquad A_0 \neq 0, \qquad A_0^2 = 0, \qquad \operatorname{supp} A = p^{-1}(\operatorname{supp} A_0).$$

Therefore the restriction

$$p : X \longrightarrow X_0 := S_0 \setminus \operatorname{supp} A_0$$

is proper. It is enough to construct a smooth psh exhaustion of $X_0$, since its pullback will be one on $X$.

By [FN05, §2], the minimal surface $S_0$ is a complex torus, a K3 surface, or a surface of class $\mathrm{VII}_0$. We first exclude the two Kähler possibilities.

Suppose $S_0$ is a torus, and choose an irreducible curve $C \subset \operatorname{supp} A_0$. A distinct translate $C'$ of $C$ is numerically equivalent to $C$, so

$$C^2 = C \cdot C' \ge 0.$$

The opposite inequality follows from [FN05, Lemma 2.1]; hence $C^2 = 0$. Since $K_{S_0}$ is trivial, adjunction gives $p_a(C) = 1$. The normalization of $C$ cannot be rational: a map $\mathbf{P}^1 \to S_0$ lifts to $\mathbb{C}^2$ and is therefore constant. Thus its normalization has genus one, and $p_a(C) = 1$ forces $C$ to be smooth. A translate of $C$ through the origin is an elliptic subtorus. The quotient by this subtorus is an elliptic curve, whose meromorphic functions pull back to nonconstant meromorphic functions on $S_0$. This contradicts $a(S_0) = 0$.

Suppose instead that $S_0$ is a K3 surface. Since $A_0$ is effective and nonzero,

$$H^0(S_0, \mathcal{O}_{S_0}(-A_0)) = 0.$$

Riemann–Roch and Serre duality therefore give

$$h^0(S_0, \mathcal{O}_{S_0}(A_0)) - h^1(S_0, \mathcal{O}_{S_0}(A_0)) = 2 + \frac{1}{2}A_0^2 = 2.$$

There are consequently two independent sections of $\mathcal{O}_{S_0}(A_0)$, whose ratio is a nonconstant meromorphic function. This again contradicts $a(S_0) = 0$.

It remains to consider $S_0$ of class $\mathrm{VII}_0$. Suppose first that $b_2(S_0) = n > 0$. Enoki's theorem [Eno81, Main Theorem, p. 454], applied to $A_0^2 = 0$, identifies $S_0$ as an Enoki surface and gives

$$A_0 = mC_{\mathrm{E}}, \qquad m \in \mathbb{Z}_{>0},$$

where $C_{\mathrm{E}}$ is its distinguished reduced rational cycle. By [Eno81, Proposition 3.5(iii), p. 460],

$$X_0 = S_0 \setminus C_{\mathrm{E}}$$

is an affine-line bundle $\mathcal{A}$ over an elliptic curve. Its linearization $Q$ has degree $-n$, with the convention of [Eno81, §2, p. 457].

Write the affine transition maps as $w_j = a_{jk}w_k + b_{jk}$. Extending them to $\mathbf{P}^1$ produces a compact ruled surface $\overline{\mathcal{A}}$ with an infinity section $Y_\infty$ and

$$\overline{\mathcal{A}} \setminus Y_\infty \simeq X_0.$$

In the coordinates $u_j = w_j^{-1}$ near $Y_\infty$,

$$u_j = \frac{u_k}{a_{jk} + b_{jk}u_k} = a_{jk}^{-1}u_k + O(u_k^2).$$

Hence

$$N_{Y_\infty/\overline{\mathcal{A}}} \simeq Q^{-1}, \qquad Y_\infty^2 = \deg Q^{-1} = n > 0.$$

Lemma 5.1(iii) gives a smooth strictly psh divergent collar along $Y_\infty$, and part (ii) gives a smooth psh exhaustion of $X_0$.

Finally, suppose that $b_2(S_0) = 0$. Since $A_0 \neq 0$, the surface $S_0$ contains a curve. The classification recalled in [Eno81, p. 453] then makes $S_0$ elliptic or Hopf. The elliptic case is excluded by $a(S_0) = 0$, so $S_0$ is Hopf.

Because $H^2(S_0, \mathbb{R}) = 0$, every irreducible curve $C$ satisfies

$$C^2 = K_{S_0} \cdot C = 0.$$

Adjunction gives $p_a(C) = 1$. Its normalization cannot be rational, since a map $\mathbf{P}^1 \to S_0$ lifts to the universal cover $\mathbb{C}^2 \setminus \{0\}$ and is constant. Thus $C$ is a smooth elliptic curve. Distinct irreducible curves have intersection number zero, so positivity of local intersection multiplicities makes them disjoint. Consequently, the connected components of $\operatorname{supp} A_0$ are disjoint smooth elliptic curves of square zero.

Let $C$ be one such component. Its normal bundle has degree zero and is therefore unitary flat. If $C$ has finite Ueda type, [CLPT18, Theorem 2.5] gives a smooth strictly psh divergent collar along $C$. If $C$ has infinite Ueda type, [CFS+26, Theorem 1.6 and Proposition 2.1] shows that $\mathcal{O}_{S_0}(C)$ is unitary flat on a neighborhood $U_C$ of $C$. For a flat Hermitian metric $h_C$ and the canonical section $s_C$, the function

$$\Phi_C = -\log \|s_C\|_{h_C}^2$$

is smooth and pluriharmonic on $U_C \setminus C$ and tends uniformly to $+\infty$ along $C$. Thus every connected boundary component has a smooth divergent psh collar. Lemma 5.1(ii) gives a smooth psh exhaustion of $X_0$.

In either class $\mathrm{VII}_0$ case, pulling this exhaustion back by the proper map $p : X \to X_0$ proves (ii). This completes the proof. □

**Theorem 5.12.** *Let $f : S \to B$ be a smooth $\mathbf{P}^1$-bundle over a compact Riemann surface and let $D \neq 0$ be a reduced SNC divisor on $S$. The three conditions in Theorem 5.11 are equivalent.*

*Proof.* As above, it suffices to construct smooth divergent collars under condition (iii). Lemma 5.1 handles boundary components with a positive eigenvalue. For a connected component $E$ with zero largest eigenvalue, choose its positive kernel divisor $A$. If $E$ is vertical, it is a whole fiber, and the punctured-base construction applies. Otherwise, for a fiber $F$ of $f$, put $a = A \cdot F > 0$. A vertical component would be a fiber and have intersection $a$ with $A$, contradicting $A \cdot E_j = 0$. Thus every $E_j$ is horizontal. Choose a section $C_0$ of the projective ruling and write $C_0^2 = -e$. Writing $\equiv$ for numerical equivalence of divisors, we have

$$A \equiv aC_0 + bF, \qquad E_j \equiv d_jC_0 + b_jF, \qquad d_j = E_j \cdot F > 0.$$

Using $C_0 \cdot F = 1$ and $F^2 = 0$, compute

$$0 = A^2 = -ea^2 + 2ab, \qquad 0 = A \cdot E_j = ab_j + d_jb - ead_j.$$

Since $a > 0$, these identities give

$$b = ae/2, \qquad E_j \equiv d_j(C_0 + (e/2)F), \qquad E_j^2 = E_j \cdot E_k = 0 \quad (j \neq k).$$

Distinct components are therefore disjoint, and connectedness makes $E = C$ a single curve, smooth by the SNC hypothesis. For $d = C \cdot F$, the canonical class formula

$$K_S \equiv -2C_0 + (2g(B) - 2 - e)F$$

and adjunction give $2g(C)-2 = (K_S+C)\cdot C = d(2g(B)-2)$. The nonconstant map $f|_C : C \to B$ is finite of degree $d$. Riemann–Hurwitz now gives degree zero for its effective ramification divisor, so it is étale.

For $d = 1$, the complement $S \setminus C$ is an affine-line bundle with degree-zero linearization. In unitary flat frames write its transitions as $w_j = a_{jk}w_k + b_{jk}(z)$ with $|a_{jk}| = 1$. The construction of [KU25, Proposition 2.1] gives pluriharmonic functions $h_j$ with $h_j = a_{jk}h_k + b_{jk}$. On simply connected charts split $h_j = f_j + g_j$, with $f_j$ holomorphic and $g_j$ antiholomorphic. The coordinate changes $w_j \mapsto w_j - f_j$ replace $b_{jk}$ by

$$c_{jk} = b_{jk} + a_{jk}f_k - f_j = g_j - a_{jk}g_k.$$

This expression is both holomorphic and antiholomorphic, hence constant. In the new coordinates,

$$w_j = a_{jk}w_k + c_{jk}, \qquad g_j = a_{jk}g_k + c_{jk}, \qquad |a_{jk}| = 1.$$

The identity $w_j - g_j = a_{jk}(w_k - g_k)$ makes $\rho = |w_j - g_j|^2$ global. Since $g_j$ is antiholomorphic,

$$\rho_{z\bar{z}} = |(g_j)_{\bar{z}}|^2, \qquad \rho_{z\bar{w}_j} = \rho_{w_j\bar{z}} = 0, \qquad \rho_{w_j\bar{w}_j} = 1.$$

Thus $\rho$ is smooth and psh. On a finite relatively compact trivializing cover of the compact base, the $g_j$ are bounded, so $\rho \leq c$ bounds each fiber coordinate. Its sublevels are compact.

For $d \geq 2$, let $B' \to B$ be the finite étale Galois cover associated to the kernel of the permutation monodromy $\pi_1(B) \to \mathfrak{S}_d$ of $C \to B$. Form

$$S' = S \times_B B', \qquad p : S' \to S, \qquad C' = p^{-1}(C) = C_1 \sqcup \cdots \sqcup C_d.$$

The $C_j$ are sections with square zero because their normal bundles are pullbacks of $N_{C/S}$ along finite covers. If $d = 2$, these sections identify the ruling with $\mathbf{P}(\mathcal{O}_{B'} \oplus L)$, where $\deg L = 0$, and $S' \setminus C' = L^\times$. Here $L^\times$ denotes the total space of $L$ with its zero section removed. For a unitary flat metric, put $t(v) = \log \|v\|^2$. Then $dd^c t = 0$ on $L^\times$, and

$$\mathcal{L}_{t^2}(\xi) = 2|\partial t(\xi)|^2 \geq 0, \qquad \{t^2 \leq c\} = \{e^{-\sqrt{c}} \leq \|v\|^2 \leq e^{\sqrt{c}}\}$$

is compact. Hence $t^2$ is a smooth psh exhaustion.

For $d \geq 3$, the unique fiberwise Möbius map sending $C_1, C_2, C_3$ to $0, 1, \infty$ identifies $S'$ with $B' \times \mathbf{P}^1$. Every remaining section is a holomorphic map $B' \to \mathbf{P}^1 \setminus \{0, 1, \infty\}$, which is constant

by the maximum principle. Thus $S' \setminus C' = B' \times (\mathbf{P}^1 \setminus Q)$ for a finite set $Q$, and an exhaustion of the punctured fiber pulls back to one of the product.

For $d \geq 2$, let $\rho' \geq 0$ be the exhaustion just constructed and $G$ the deck group of $B' \to B$. The function

$$\widehat{\rho} = \frac{1}{|G|} \sum_{g \in G} \rho' \circ g$$

is smooth, psh, and $G$-invariant. Since $\widehat{\rho} \geq \rho'/|G|$, its sublevels are compact. It descends under the finite étale map $p$ to a smooth psh exhaustion of $S \setminus C$. Restrict this function to a collar at $E$ and apply Lemma 5.1(ii) to glue all boundary collars. □

5.4. **Square-zero curves.** For a smooth square-zero curve, Ueda theory relates the formal neighborhood to the existence of holomorphic defining functions $w_j = t_{jk} w_k$ with $|t_{jk}| = 1$. These functions give a pluriharmonic collar through $-\log |w_j|^2$.

**Proposition 5.13.** *Let $C \subset S$ be smooth, connected, and square-zero, with $S$ compact Kähler. Put $N = N_{C/S}$, let $\chi : H_1(C, \mathbb{Z}) \to \mathrm{U}(1)$ be its unitary character, and set $K = \ker[H_1(C, \mathbb{Z}) \to H_1(S, \mathbb{Z})]$. The complement $S \setminus C$ has a smooth psh exhaustion if one of the following conditions holds.*

(i) *$C$ has finite Ueda type or a unitary flat neighborhood.*
(ii) *$N$ is torsion or Diophantine.*
(iii) *$\chi(K)$ is finite.*
(iv) *$q(S) \geq g(C)$.*
(v) *$g(C) = 1$.*

*Here Diophantine means $\operatorname{dist}(N^{\otimes m}, \mathcal{O}_C) \geq c m^{-\tau}$ for some $c, \tau > 0$ and all $m \geq 1$, with respect to a fixed flat metric on $\mathrm{Pic}^0(C)$. We write $q(S) = h^1(S, \mathcal{O}_S)$ for the irregularity.*

*Proof.* For condition (i), finite type gives a smooth strictly psh collar [CLPT18, Theorem 2.5]. In a unitary flat neighborhood, $w_j = t_{jk} w_k$ with $|t_{jk}| = 1$, so the functions $-\log |w_j|^2$ agree and form a pluriharmonic divergent collar. And for condition (ii) in infinite type, Ueda's convergence theorem [Ued83, Theorem 3] supplies such a neighborhood when $N$ is torsion or Diophantine.

For condition (iii), suppose $\chi(K)$ is finite, and let $k$ annihilate this finite group. Write $j_* : H_1(C, \mathbb{Z}) \to H_1(S, \mathbb{Z})$ for the inclusion map. The formula

$$\chi_0(j_* \gamma) = \chi(\gamma)^k$$

defines a character on $\operatorname{im} j_*$, because two lifts differ by an element of $K$. Divisibility of $\mathrm{U}(1)$ extends it to a character $\widetilde{\chi} : H_1(S, \mathbb{Z}) \to \mathrm{U}(1)$, with $\widetilde{\chi} \circ j_* = \chi^k$. For non-torsion $N$ of infinite type, [CLPT18, Theorem 4.4] then gives an effective divisor $E$ disjoint from $C$ with $[E] = m[C]$ for some $m > 0$. The line bundle $\mathcal{O}_S(mC - E)$ has zero real Chern class and hence a unitary flat metric on the compact Kähler surface. Its canonical meromorphic section $\sigma = s_C^m / s_E$ satisfies

$$\operatorname{div}(\sigma) = mC - E, \qquad dd^c \log \|\sigma\|^2 = m[C] - [E].$$

On a neighborhood of $C$ disjoint from $E$, the function $-m^{-1} \log \|\sigma\|^2$ is therefore pluriharmonic off $C$ and equals $-\log |w|^2$ plus a smooth function. It is the required divergent collar.

For condition (iv), consider the restriction $r : H^1(S, \mathcal{O}_S) \to H^1(C, \mathcal{O}_C)$. If $r$ is not injective, [CLPT18, Proposition 2.7] makes a multiple of $C$ a fiber, and the punctured-base construction applies. Otherwise, since $q(S) \geq g(C)$, injectivity makes $r$ an isomorphism. Hodge decomposition and complex conjugation then give an isomorphism $H^1(S, \mathbb{Q}) \to H^1(C, \mathbb{Q})$, whose dual is $j_* \otimes \mathbb{Q}$. Consequently $K \otimes \mathbb{Q} = 0$. As a subgroup of the free abelian group $H_1(C, \mathbb{Z})$, $K$ is torsion-free, hence $K = 0$, and the preceding character argument applies.

Finally, for (v), [CFS+26, Theorem 1.6] shows that an elliptic curve of infinite Ueda type has a unitary flat neighborhood. We conclude by applying Lemma 5.1(ii) in all cases. □

## 6. Holomorphic convexity over Stein spaces

Throughout this section complex spaces are reduced, Hausdorff, and countable at infinity. All statements about fiber components concern their reduced supports. For a continuous psh exhaustion $\rho$ on $X$, we write $X(t) = \{x \in X : \rho(x) < t\}$.

**Theorem 6.1** (=Theorem 1.4)**.** *Let $X$ be a reduced complex space admitting a continuous psh exhaustion, and let $F : X \to S$ be a holomorphic map to a Stein space. Suppose that every irreducible component of every reduced fiber of $F$ is either compact or Stein. Then $X$ is holomorphically convex.*

The theorem allows fiber components of arbitrary dimension and imposes no purity assumption on $X$. When $X$ is finite-dimensional, Remark 5.3 also gives weak 1-completeness. The key step in the proof is to choose an exhaustion level avoiding compact components of a fixed fiber and place the part of that level over a small base neighborhood inside a Stein open set. We then apply [Vîj22b, Corollary 2.5] to obtain local holomorphic convexity. The choice of levels and the use of Stein neighborhoods also occur in [KR98, Lemma 3.1].

*Proof.* Let $\rho$ be a continuous psh exhaustion of $X$. Fix $\lambda \in \mathbb{R}$. We first show that, for every $s \in S$, there are $c > \lambda$ and a Stein open neighborhood $B$ of $s$ such that

$$D = F^{-1}(B) \cap X(c)$$

is holomorphically convex.

**Step I: Choice of level sets.** Fix $s \in S$. By the maximum principle, $\rho$ is constant on every compact irreducible component of $F^{-1}(s)_{\mathrm{red}}$. The components form a locally finite family, so only finitely many meet any compact sublevel $\{\rho \le R\}$. Consequently, only finitely many of these constant values lie below any given bound. Choose $c > \lambda$ avoiding them. Then

$$L = F^{-1}(s) \cap \{\rho = c\}$$

meets no compact irreducible component of the central fiber.

**Step II: Local holomorphic convexity.** If $L = \varnothing$, choose a Stein neighborhood $B$ of $s$ disjoint from the compact set $F(\{\rho = c\})$. For each compact $T \subset B$,

$$(F|_D)^{-1}(T) = F^{-1}(T) \cap \{\rho \le c\}$$

is compact. Thus $D \to B$ is proper and $D$ is holomorphically convex.

Suppose $L \neq \varnothing$, and let $Y_s$ be the union of all noncompact irreducible components of $F^{-1}(s)_{\mathrm{red}}$. By local finiteness, $Y_s$ is a closed analytic subspace of $X$. Its irreducible components are Stein, so $Y_s$ is Stein by the corollary following [Nar62, Theorem 1, p. 327]. Siu's theorem [Siu76] gives a Stein open neighborhood $N \supset Y_s$ in $X$. Since $L \subset Y_s$, the compact set $\{\rho = c\} \setminus N$ has image under $F$ disjoint from $s$. We may therefore choose a Stein open neighborhood $B$ of $s$ with

$$F^{-1}(B) \cap \{\rho = c\} \subset N.$$

Put $M = F^{-1}(B)$ and $D = M \cap X(c)$. We check the hypotheses of [Vîj22b, Corollary 2.5] for $D \subset M$. First, if $\sigma_B \ge 0$ is a continuous psh exhaustion of $B$, then

$$\eta = \max\left\{\frac{1}{c-\rho},\, \sigma_B \circ F\right\}$$

is a continuous psh exhaustion of $D$. Indeed, for $t > 0$,

$$\{\eta \le t\} = \left\{\rho \le c - \frac{1}{t}\right\} \cap F^{-1}(\{\sigma_B \le t\})$$

is compact and contained in $D$. Next, take the closure and boundary relative to $M$. For every compact $T \subset B$, the set $\overline{D}^M \cap F^{-1}(T)$ is closed in the compact set $\{\rho \le c\} \cap F^{-1}(T)$. Hence $\overline{D}^M$ is proper over $B$. Moreover,

$$\partial_M D \subset M \cap \{\rho = c\} \subset N \cap M.$$

For each $t \in B$, the fiber of $N \cap M \to B$ is $N \cap F^{-1}(t)$, a closed analytic subspace of the Stein space $N$, hence Stein. The cited corollary now gives the holomorphic convexity of $D$.

In either case, [Vîj22b, Proposition 2.2], applied to the continuous psh function $\rho|_D$, shows that

$$F^{-1}(B) \cap X(\lambda) = \{x \in D : \rho(x) < \lambda\}$$

is holomorphically convex.

**Step III: Passage to $X$.** Thus every $s \in S$ has a neighborhood whose inverse image in $X(\lambda)$ is holomorphically convex. Since $(\lambda - \rho)^{-1}$ is a continuous psh exhaustion of $X(\lambda)$, [Vîj22b, Proposition 2.4] implies that $X(\lambda)$ is holomorphically convex. That proposition uses the relative Remmert reduction and the Steinness argument based on [Vâj06, Lemma 1, p. 541] and Siu's theorem. Finally, $\lambda$ was arbitrary, so [Vîj22b, Proposition 2.2] gives the holomorphic convexity of $X$. □

We first deduce a consequence formulated in terms of local fiber dimension, in the spirit of Ohsawa's surface criterion. It will also give a direct deduction of Vîjîitu's rank criterion.

**Corollary 6.2.** *Let $X$ be a pure $n$-dimensional complex space, $n \ge 2$, admitting a continuous psh exhaustion. Suppose that there exist a holomorphic map $F : X \to \mathbb{C}^{n-1}$, a compact subset $K_0 \subset X$, and a closed analytic subset $E \subset X$, each of whose connected components is compact, such that*

$$\dim_x F^{-1}(F(x)) \le 1 \qquad \text{for every } x \in X \setminus (K_0 \cup E).$$

*Then $X$ is holomorphically convex.*

*Proof.* Every irreducible component of a nonempty fiber has dimension at least one, since the fiber is defined by $n-1$ holomorphic functions on a pure $n$-dimensional space. Let $C$ be a fiber component with $\dim C > 1$. The dimension hypothesis implies $C \subset K_0 \cup E$. If $C \subset E$, then $C$ lies in a compact connected component of $E$ and is compact. Otherwise $C \cap E$ is a proper analytic subset of $C$, so $C \setminus E$ is dense in $C$. Since $C \setminus E \subset K_0$ and $K_0$ is closed, $C \subset K_0$ and $C$ is again compact. Every remaining noncompact component is a curve and hence Stein [Nar62, p. 327]. Therefore, Theorem 6.1 applies. □

We next recover [Vîj22b, Theorem 1.1]. For a smooth point $x$ of a pure $n$-dimensional space $X$, let

$$\mathcal{H}_x = \{y \in X : f(y) = f(x) \text{ for every } f \in \mathcal{O}(X)\}, \qquad \operatorname{rank}_x \mathcal{O}(X) = n - \dim_x \mathcal{H}_x.$$

**Corollary 6.3** (Vîjîitu, [Vîj22b, Theorem 1.1])**.** *Let $X$ be a pure $n$-dimensional complex space, $n \ge 2$, admitting a continuous psh exhaustion. Suppose that every connected component of $X_{\text{sing}}$ is compact and that there is a compact subset $K \subset X$ such that*

$$\operatorname{rank}_x \mathcal{O}(X) = n - 1 \qquad \text{for every } x \in X_{\text{reg}} \setminus K.$$

*Then $X$ is holomorphically convex.*

*Proof.* Put $\Omega = X_{\mathrm{reg}} \setminus K$. If $\Omega = \varnothing$, the density of $X_{\mathrm{reg}}$ in $X$ gives $X \subset K$, so $X$ is compact and the conclusion follows. Otherwise consider the continuous restriction map

$$r : \mathcal{O}(X) \longrightarrow \mathcal{O}(\Omega), \qquad \mathcal{A} = \mathcal{O}(X)/\ker r.$$

The kernel is closed, so $\mathcal{A}$ is a Fréchet space. We identify it with the image of $r$, equipped with the quotient topology; its inclusion into $\mathcal{O}(\Omega)$ is continuous. The germ at each $x \in \Omega$ of the common level set of $\mathcal{A}$ is the same as for $\mathcal{O}(X)$. Hence

$$\operatorname{rank}_x \mathcal{A} = n - 1 \qquad (x \in \Omega).$$

By [Vîj22b, Theorem 3.1], there exist $a_1, \dots, a_{n-1} \in \mathcal{A}$ such that

$$(a_1, \dots, a_{n-1}) : \Omega \longrightarrow \mathbb{C}^{n-1}$$

has pure one-dimensional nonempty fibers. Choose representatives $f_1, \dots, f_{n-1} \in \mathcal{O}(X)$ and set $F = (f_1, \dots, f_{n-1}) : X \to \mathbb{C}^{n-1}$. Since $\Omega$ is open in $X$,

$$\dim_x F^{-1}(F(x)) = 1 \qquad (x \in \Omega).$$

Corollary 6.2 applies with $K_0 = K$ and $E = X_{\mathrm{sing}}$. □

*Remark* 6.4. The two products below satisfy Theorem 6.1 but fail the hypotheses of Corollaries 6.3 and 6.2, respectively. Let

$$C = \{[u : v : w] \in \mathbf{P}^2 : v^2 w = u^2(u + w)\}$$

be the irreducible projective nodal cubic with singular point $p = [0 : 0 : 1]$. Set

$$X = C \times \mathbb{C}^2, \qquad F = \mathrm{pr}_2, \qquad \rho(c, z) = \|z\|^2.$$

Every fiber is compact, so Theorem 6.1 applies. Every holomorphic function on $X$ is constant on the fibers and descends to $\mathbb{C}^2$. Thus

$$\mathcal{O}(X) = F^*\mathcal{O}(\mathbb{C}^2), \qquad \operatorname{rank}_x \mathcal{O}(X) = 2 \quad (x \in X_{\mathrm{reg}}).$$

However, $X_{\mathrm{sing}} = \{p\} \times \mathbb{C}^2$ is connected and noncompact. Hence the singular-locus hypothesis of [Vîj22b, Theorem 1.1] fails even though its rank condition holds.

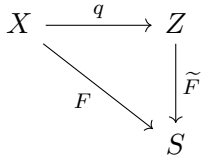
For a different distinction, take

$$X = \mathbf{P}^2 \times \mathbb{C}, \qquad F = \mathrm{pr}_2, \qquad \rho(p, z) = |z|^2.$$

Theorem 6.1 again applies. Every holomorphic map $G : X \to \mathbb{C}^2$ is constant on each $\mathbf{P}^2 \times \{z\}$, so $\dim_x G^{-1}(G(x)) \geq 2$ everywhere. The condition in Corollary 6.2 would therefore require $X = K_0 \cup E$. If $E$ is proper, its complement is dense in the irreducible space $X$, forcing $X \subset K_0$; if $E = X$, its connected component is noncompact. Both alternatives contradict the hypotheses of that corollary. Also $\operatorname{rank}_x \mathcal{O}(X) = 1$, so [Vîj22b, Theorem 1.1] does not apply. Both products are already holomorphically convex by properness of the projection; they illustrate the difference between the hypotheses.

We finish with a description of the Remmert reduction under the general hypotheses.

**Corollary 6.5.** *Under the hypotheses of Theorem 6.1, let $q : X \to Z$ be the Remmert reduction. There is a holomorphic map $\widetilde{F} : Z \to S$ making the following diagram commute:*




*The positive-dimensional fibers of $q$, with their reduced supports, are exactly the maximal connected unions of positive-dimensional compact irreducible components of a fixed fiber of $F$. Every such union is finite.*

*Proof.* Since $S$ is Stein, its universal property gives a holomorphic map $\widetilde{F} : Z \to S$ with $F = \widetilde{F} \circ q$. In particular, every $q$-fiber lies in an $F$-fiber. Also, since $Z$ is Stein, $q$ contracts every compact irreducible analytic subset of $X$ to a point.

Let $T$ be a connected union of positive-dimensional compact irreducible components of a fixed $F$-fiber. Local finiteness and connectedness imply that any two components of $T$ can be joined by a finite chain of intersecting components. Their images under $q$ therefore coincide, so $T$ lies in a single $q$-fiber $Q$. This fiber is compact and hence meets only finitely many irreducible components of the $F$-fiber. Thus $T$ is a finite union.

Conversely, let $Q$ be a positive-dimensional $q$-fiber, with its reduced structure. It is compact and connected, and each of its irreducible components has positive dimension: a zero-dimensional component would be an isolated point. Let $A$ be an irreducible component of $Q$, and choose an irreducible component $C$ of the containing $F$-fiber with $A \subset C$. If $C$ were noncompact, it would be Stein and could not contain the positive-dimensional compact analytic subset $A$. Hence $C$ is compact. Since $q$ contracts $C$ and $C$ meets $Q$, we have $C \subset Q$. The maximality of $A$ as an irreducible component of $Q$ now gives $A = C$. Thus $Q$ is itself a finite connected union of positive-dimensional compact irreducible components of the $F$-fiber.

Every union $T$ considered above is therefore contained in a positive-dimensional $q$-fiber which is itself such a union. Moreover, any such connected union meeting a $q$-fiber lies entirely in that fiber. These fibers are consequently exactly the maximal connected unions in the statement. □

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